

Full Length Article

Learning-based local RBF neural network with adaptively supported domain for solving partial differential equations

Zhenzhou Fan^a, Min Lei^{a,*}, Ruiping Niu^a, Letian Wang^a, Hongen Jia^{a,b}, Peiyao Liu^a^a College of Mathematics, Taiyuan University of Technology, China^b Shanxi Key Laboratory for Intelligent Optimization Computing and Blockchain Technology, China

ARTICLE INFO

Keywords:

Local RBF

Adaptively supported domain

Neural network

Numerical stability

ABSTRACT

The Local Radial Basis Function method (Local RBF) is attractive for its sparsity and efficiency. However, its accuracy and stability depend critically on the choice of the local support domain. Conventional methods rely on trial-and-error to select a uniform support domain size for all center points, lack theoretical guidance, and often exhibit significant sensitivity of numerical error to this choice. This paper introduces a learning-based Local RBF Neural Network with Adaptively Supported Domain (Local RBF-ASD), which mitigates this sensitivity by adaptively adjusting the effective support domain within a differentiable training loop. Both the model parameters and unknown field quantities are optimized through a composite loss function that combines interior PDE residuals with boundary constraints, while preserving the sparse structure of the discretization. Numerical benchmarks on (i) Helmholtz problems with complex geometries (including anisotropic variants), (ii) steady Stokes flows, and (iii) a three-dimensional near-singular Poisson problem demonstrate that Local RBF-ASD achieves accurate and stable solutions across a wide range of support domain sizes.

1. Introduction

Partial differential equations (PDEs) provide a fundamental mathematical framework for describing diverse physical phenomena in science and engineering. Over the past decades, various mesh-based numerical methods, such as the finite difference method (LeVeque, 2007), finite element method (FEM) (Zienkiewicz et al., 2005), and finite volume method (Versteeg, 2007), have been developed to obtain accurate and efficient PDE solutions. However, these approaches rely on structured or unstructured meshes whose generation and adaptation are often complex and computationally expensive, particularly for nonlinear, high-dimensional, or moving-boundary problems (Kundu et al., 2024).

To overcome the limitations of mesh generation, a broad class of meshfree methods has been developed (Belytschko et al., 1996). Early examples include smoothed particle hydrodynamics (SPH) (Gingold & Monaghan, 1977; Lucy, 1977) and moving least squares-based methods such as the diffuse element (Nayroles et al., 1992) and element-free Galerkin (Belytschko et al., 1994) formulations. The global radial basis function (RBF) collocation method (Kansa, 1990a,b) subsequently emerged as a strong-form, meshfree discretization, where PDEs are collocated at interior nodes and boundary conditions are enforced using the same RBF trial space. Although the approach can achieve spectral or near-spectral convergence with a properly chosen shape parameter,

it leads to dense interpolation matrices that hinder large-scale applications (Kansa, 1990b). Boundary-type RBF formulations, such as the method of fundamental solutions (MFS) (Kupradze & Aleksidze, 1964; Mathon & Johnston, 1977) and the boundary knot method (BKM) (Chen & Tanaka, 2002), further reduce dimensionality by enforcing collocation only on the boundary, with BKM eliminating the singular kernels inherent to MFS. Compared with these boundary-type schemes, the global RBF method exhibits broader applicability, particularly for problems lacking fundamental or general solutions, such as those with variable coefficients or nonlinearities. To mitigate the computational burden of global RBF formulations, researchers have developed strategies to sparsify the system matrices. Two main approaches have proven effective: the use of compactly supported RBFs (Wendland, 2005; Zhang et al., 2004), which naturally yield sparse matrices, and the Local RBF collocation method (Chandhini & Sanyasiraju, 2010; Shu et al., 2005), which restricts the computational stencil to neighboring nodes. These localization strategies substantially reduce the computational complexity from $\mathcal{O}(N^3)$ to tractable levels while retaining favorable approximation properties (Qiao et al., 2017).

Despite their computational advantages, Local RBF methods introduce additional challenges in parameter selection. Their numerical performance depends strongly on two quantities: the shape parameter of RBF kernels, such as the multiquadric (MQ) function, and the

* Corresponding author.

E-mail address: leimin@tyut.edu.cn (M. Lei).<https://doi.org/10.1016/j.neunet.2026.109643>

Received 18 April 2026; Received in revised form 1 August 2026; Accepted 12 September 2026

Available online 15 September 2026

0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

number of nodes in each local support domain (Rippa, 1999). These parameters are commonly selected using heuristic rules or trial-and-error procedures, which lack general theoretical guidance and may produce poorly conditioned local systems or degraded approximations. Considerable effort has been devoted to adaptive shape-parameter selection. For example, an adaptive shape-parameter strategy for Local RBF discretizations was proposed in Li et al. (2023), yielding improved numerical results and computational efficiency. In contrast, the selection of the local support-domain size, also referred to as the stencil size, remains largely problem dependent. Sarra (2023) reported that two-dimensional applications commonly employ between 20 and 100 nodes, but no generally applicable selection criterion is available. Consequently, a robust mechanism for determining effective local interactions is desirable to reduce repeated problem-specific tuning.

Conventional procedures for determining the local stencil size, including trial-and-error, grid search, and other hyperparameter optimization strategies, treat N_s as an external discrete parameter. Each candidate value requires an independent construction and solution of the Local RBF system, after which one value is generally applied uniformly to all local centers. In benchmark studies, the selection is often based on comparison with the exact solution, whereas such information is unavailable in most practical PDE applications. Moreover, a globally fixed value cannot account for spatial variations in node distribution, geometry, boundary proximity, or local solution behavior.

The proposed Local RBF-ASD method addresses this limitation by introducing a center-dependent and differentiable gating parameter for each local stencil. For a prescribed candidate neighborhood, the gates are optimized jointly with the numerical solution through the PDE-constrained loss, allowing the effective interaction pattern to vary among local centers. Thus, N_s specifies the maximum candidate neighborhood, whereas the learned gates determine the effective contributions of the nodes within that neighborhood. The method therefore enables spatially nonuniform support configurations and reduces the dependence on a manually selected global stencil size.

Recent advances in machine learning have opened new avenues for solving PDEs and automating parameter selection in numerical methods. Physics-informed neural networks (PINNs) (Raissi et al., 2019) embed governing PDEs directly into the loss function as physical constraints, enabling neural networks to approximate solutions that respect underlying physics laws. Rigorous error analyses have been developed for PINNs applied to dynamic hyperbolic PDEs of second order in time, establishing relationships among approximation error, training loss, and data distribution, and motivating improved formulations involving initial and boundary residuals (Qian et al., 2023). However, due to the frequency principle (Zhi-Qin et al., 2020), standard PINNs often still struggle with high-frequency problems, as low-frequency components are fitted preferentially during training. To address such limitations, hybrid approaches combining machine learning with traditional numerical methods have shown promise. For instance, the random feature method provides a unified framework bridging machine learning and classical algorithms for PDEs solving (Chen et al., 2022). A learning-based numerical method for high-frequency Helmholtz equations combines fundamental solutions with Tikhonov regularization and provides theoretical error estimates based on quantitative Runge approximation (Chen et al., 2025). Furthermore, a physics-informed radial basis network (PIRBN) employs a shallow radial basis function architecture to maintain locality during training, proving more effective and efficient than conventional PINNs for nonlinear PDEs involving high-frequency features or ill-posed domains (Bai et al., 2023). These developments underscore the potential of integrating traditional numerical methods with machine learning paradigms. In particular, learning-based strategies offer new approaches to longstanding challenges in classical methods, such as adaptive parameter selection. An adaptive method of fundamental solutions employs a PINN architecture to optimize source-point locations and reduce manual tuning (Wang et al., 2025). Similarly, a neural formulation of the local method of fundamental solutions determines source radii for indi-

Table 1

Some commonly used RBFs.

RBFs Name	$\phi(r; c_s)$
Gaussian	$\exp(-(c_s r)^2)$
Multiquadric (MQ)	$(1 + (c_s r)^2)^{1/2}$
Inverse MQ (IMQ)	$(1 + (c_s r)^2)^{-1/2}$

vidual subdomains, replacing repeated trial-and-error procedures with learnable parameter adaptation (Han et al., 2025).

Motivated by these developments, this study proposes a learning-based Local RBF Neural Network with Adaptively Supported Domain, referred to as Local RBF-ASD. The method introduces trainable gating parameters into the local RBF matrices and jointly optimizes these parameters with the unknown nodal values. The resulting framework preserves the local structure of the discretization while allowing the effective interactions to vary among centers. It therefore reduces the dependence on a manually prescribed stencil size rather than completely eliminating the candidate neighborhood parameter. Numerical experiments involving Helmholtz equations, steady Stokes flows, and a three-dimensional near-singular Poisson problem demonstrate consistent performance over a broad range of candidate stencil sizes.

The remainder of this paper is organized as follows: Section 2 provides a detailed overview of the theoretical foundations of the Local RBF collocation method. Section 3 presents the Local RBF-ASD method, including the implementation of the adaptively supported domain mechanism and the optimization strategies employed in the network. In Section 4, we present a series of numerical experiments to demonstrate the performance of the Local RBF-ASD method, comparing it to the conventional Local RBF method. Finally, Section 5 concludes the paper with a discussion of the results and potential future directions for this approach.

2. Local RBF collocation method

Radial basis functions (RBFs) form an important class of real-valued functions, $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$, characterized by their dependence solely on the Euclidean distance from a center point. Formally, these functions can be expressed as:

$$\phi(\mathbf{x}) = \phi(\|\mathbf{x} - \mathbf{x}_c\|; c_s) = \phi(r; c_s), \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^d$ denotes the input point, $\mathbf{x}_c \in \mathbb{R}^d$ is the center, $r = \|\mathbf{x} - \mathbf{x}_c\|_2$ represents their Euclidean distance, and $c_s > 0$ is the shape parameter. We consider three commonly used infinitely smooth RBFs, as summarized in Table 1.

The shape parameter $c_s > 0$ governs the flatness and radial scaling of the basis functions. All RBFs are radially symmetric with respect to the center $\mathbf{x}_c$. The Gaussian and IMQ functions exhibit monotonic decay with increasing radial distance, whereas the MQ function increases smoothly. Variations in c_s primarily affect the radial extent and localization of the RBFs, thereby influencing the smoothness and numerical conditioning of the resulting approximation.

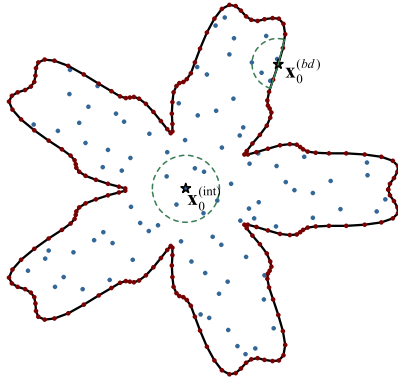
Consider a boundary value problem defined on a domain $\Omega \subseteq \mathbb{R}^d$:

$$\mathcal{L}u(\mathbf{x}) = f(\mathbf{x}), \quad \mathbf{x} \in \Omega, \quad (2)$$

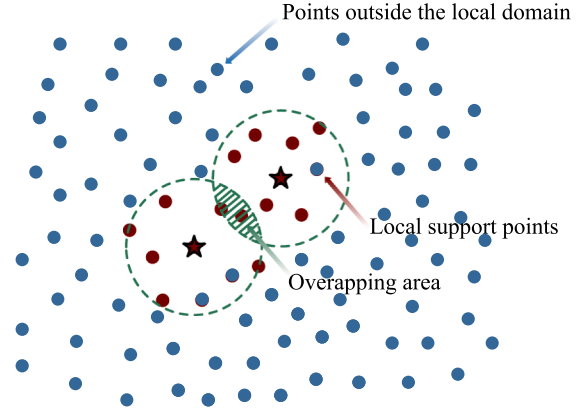
$$Bu(\mathbf{x}) = h(\mathbf{x}), \quad \mathbf{x} \in \partial\Omega, \quad (3)$$

where $\mathcal{L}$ denotes the differential operator, u is the unknown field variable, f is the source term, $\partial\Omega$ is the domain boundary, and B denotes the boundary operator.

The global RBF method results in dense coefficient matrices, leading to a computational complexity of $\mathcal{O}(N^3)$ and posing significant challenges for large-scale problems. To mitigate this issue, Local RBF method has been developed. These approaches employ only a small set of neighboring nodes for each collocation point, substantially lowering computational cost while preserving accuracy.



(a) Collocation points distribution



(b) Local support domains

Fig. 1. Illustration of the local RBF method.

For each central collocation point $\mathbf{x}_0^{(s)} := \mathbf{x}_s$, with $s = 1, 2, \dots, N$, we define $\mathbf{x}_s$ -related local subdomain $\Omega_s \subset \Omega$ and its associated node set $\mathcal{X}^{(s)}$, which consists of the k nearest neighbors together with the central point itself:

$$\mathcal{X}^{(s)} = \{\mathbf{x}_0^{(s)}, \mathbf{x}_1^{(s)}, \mathbf{x}_2^{(s)}, \dots, \mathbf{x}_k^{(s)}\} \subset \Omega_s. \quad (4)$$

Here, N denotes the total number of collocation points, s is the index of the central point, and $N_s = k + 1$ denotes the number of nodes in the candidate local stencil associated with $\mathbf{x}_s$. The stencil contains the center $\mathbf{x}_0^{(s)} = \mathbf{x}_s$ and its k nearest neighbors. In this work, the same candidate size N_s is used for all centers, with $N_s \ll N$ (Fig. 1).

Within each local subdomain Ω_s , the approximation function is expressed as a linear combination of RBFs:

$$\bar{u}(\mathbf{x}) = \sum_{j=0}^k \alpha_j^{(s)} \phi(\|\mathbf{x} - \mathbf{x}_j^{(s)}\|), \mathbf{x} \in \Omega_s \quad (5)$$

where $\alpha_j^{(s)}$ are the local expansion coefficients to be determined and $\phi(\cdot)$ is the chosen radial basis function.

Applying the collocation condition to the nodes in Ω_s yields the local interpolation system:

$$\mathbf{A}^{(s)} \boldsymbol{\alpha}^{(s)} = \mathbf{u}^{(s)}, \quad s = 1, 2, \dots, N, \quad (6)$$

where $\boldsymbol{\alpha}^{(s)} = [\alpha_0^{(s)}, \alpha_1^{(s)}, \dots, \alpha_k^{(s)}]^\top \in \mathbb{R}^{N_s}$ is the vector of local coefficients, $\mathbf{u}^{(s)} = [u(\mathbf{x}_0^{(s)}), u(\mathbf{x}_1^{(s)}), \dots, u(\mathbf{x}_k^{(s)})]^\top \in \mathbb{R}^{N_s}$ contains the nodal function values, and $\mathbf{A}^{(s)} \in \mathbb{R}^{N_s \times N_s}$ is the local interpolation matrix with entries

$$A_{ij}^{(s)} = \phi(\|\mathbf{x}_i^{(s)} - \mathbf{x}_j^{(s)}\|), \quad i, j = 0, 1, \dots, k. \quad (7)$$

The local coefficients are then formally given by

$$\boldsymbol{\alpha}^{(s)} = (\mathbf{A}^{(s)})^{-1} \mathbf{u}^{(s)}. \quad (8)$$

For an interior point $\mathbf{x}_0^{(s)} \in \mathcal{X}_{\text{int}}$, applying the differential operator $\mathcal{L}$ to the local approximation Eq. (5) yields

$$\mathcal{L}\bar{u}(\mathbf{x}_0^{(s)}) = \sum_{j=0}^k \alpha_j^{(s)} \mathcal{L}\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_j^{(s)}\|), \quad (9)$$

which satisfies

$$\mathcal{L}\bar{u}(\mathbf{x}_0^{(s)}) = f(\mathbf{x}_0^{(s)}). \quad (10)$$

Similarly, for a boundary point $\mathbf{x}_0^{(s)} \in \mathcal{X}_{\text{bd}}$, applying the boundary operator B yields

$$B\bar{u}(\mathbf{x}_0^{(s)}) = \sum_{j=0}^k \alpha_j^{(s)} B\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_j^{(s)}\|), \quad (11)$$

which satisfies

$$B\bar{u}(\mathbf{x}_0^{(s)}) = h(\mathbf{x}_0^{(s)}). \quad (12)$$

Eq. (9), and Eq. (11) can be written in vector form as

$$\mathcal{L}\Phi^{(s)} \boldsymbol{\alpha}^{(s)} = f(\mathbf{x}_0^{(s)}), \quad (13)$$

$$B\Phi^{(s)} \boldsymbol{\alpha}^{(s)} = h(\mathbf{x}_0^{(s)}), \quad (14)$$

where $\mathcal{L}\Phi^{(s)}$ and $B\Phi^{(s)}$ are $1 \times N_s$ row vectors defined by

$$\mathcal{L}\Phi^{(s)} = [\mathcal{L}\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_0^{(s)}\|), \mathcal{L}\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_1^{(s)}\|), \dots, \mathcal{L}\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_k^{(s)}\|)], \quad (15)$$

and

$$B\Phi^{(s)} = [B\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_0^{(s)}\|), B\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_1^{(s)}\|), \dots, B\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_k^{(s)}\|)]. \quad (16)$$

Substituting Eq. (8) into Eqs. (13), (14) leads to

$$\mathcal{L}\Phi^{(s)} (\mathbf{A}^{(s)})^{-1} \mathbf{u}^{(s)} = f(\mathbf{x}_0^{(s)}), \quad (17)$$

$$B\Phi^{(s)} (\mathbf{A}^{(s)})^{-1} \mathbf{u}^{(s)} = h(\mathbf{x}_0^{(s)}). \quad (18)$$

We now define the combined operator vector $\Theta^{(s)} \in \mathbb{R}^{1 \times N_s}$ as

$$\Theta^{(s)} = \begin{cases} \mathcal{L}\Phi^{(s)} (\mathbf{A}^{(s)})^{-1}, & \text{if } \mathbf{x}_0^{(s)} \in \mathcal{X}_{\text{int}}, \\ B\Phi^{(s)} (\mathbf{A}^{(s)})^{-1}, & \text{if } \mathbf{x}_0^{(s)} \in \mathcal{X}_{\text{bd}}. \end{cases} \quad (19)$$

This allows us to write the local collocation condition in the unified form

$$\Theta^{(s)} \mathbf{u}^{(s)} = g(\mathbf{x}_0^{(s)}), \quad (20)$$

where $g(\mathbf{x}_0^{(s)}) = f(\mathbf{x}_0^{(s)})$ for interior points and $g(\mathbf{x}_0^{(s)}) = h(\mathbf{x}_0^{(s)})$ for boundary points.

The global sparse system is constructed by mapping local contributions to their corresponding global positions. Let $\mathbf{I}_s = [i_0^{(s)}, i_1^{(s)}, \dots, i_k^{(s)}]$ denote the global index set for the local nodes in $\mathcal{X}^{(s)}$, where $i_j^{(s)}$ corresponds to the global index of node $\mathbf{x}_j^{(s)}$. Assembling all local subdomains yields the discrete collocation system:

$$\bar{\Theta} \mathbf{u} = \mathbf{g}, \quad (21)$$

where $\bar{\Theta} \in \mathbb{R}^{N \times N}$ is the global coefficient matrix, $\mathbf{g} = [g(\mathbf{x}_1), g(\mathbf{x}_2), \dots, g(\mathbf{x}_N)]^\top \in \mathbb{R}^N$ is the right-hand side vector assembled from local contributions, and $\mathbf{u} = [u(\mathbf{x}_1), u(\mathbf{x}_2), \dots, u(\mathbf{x}_N)]^\top \in \mathbb{R}^N$ is the global solution vector.

The s th row of $\tilde{\Theta}$ is populated by assigning elements of $\Theta^{(s)}$ to columns corresponding to global indices in $\mathbf{I}_s$:

$$\tilde{\Theta}_{s,j} = \begin{cases} \Theta_{\ell}^{(s)}, & \text{if } j = i_{\ell}^{(s)}, \ell \in \{0, 1, \dots, k\}, \\ 0, & \text{otherwise,} \end{cases} \quad j = 1, \dots, N. \quad (22)$$

Each row of $\tilde{\Theta}$ thus corresponds to the application of the relevant differential operator at a collocation point using its local subdomain. The resulting global matrix is sparse, with each row containing at most N_s non-zero entries. This structure considerably reduces both storage and computational cost relative to the dense global RBF approach. The sparse system Eq. (21) can be efficiently solved using iterative methods or direct sparse solvers, making the Local RBF method well-suited for large-scale numerical simulations.

3. Local RBF neural network with adaptively supported domain

A fundamental challenge in the Local RBF method concerns the optimal selection of local support domain sizes N_s . Conventional approaches typically rely on heuristic strategies or empirical trial-and-error procedures, which lack rigorous theoretical justification and often lead to numerical instability due to the sensitivity of approximation accuracy to the choice of support domain size. To address these limitations, we propose a novel learning-based framework that incorporates a gating mechanism with learnable parameters $b_R^{(s)}$. This network adaptively adjusts local support domains through the minimization of a composite loss function, thereby reducing the sensitivity of numerical accuracy to the selection of N_s and ensuring stable computational results. The proposed method maintains full differentiability throughout the training process while enabling dynamic optimization of support domain configurations.

3.1. Network structure

For clarity, the principal variables used in the proposed framework are summarized here. Let $\mathcal{X} = \{\mathbf{x}_i\}_{i=1}^N = \mathcal{X}_{\text{int}} \cup \mathcal{X}_{\text{bd}}$ denote the complete set of N collocation points. For each center $\mathbf{x}_s$, N_s denotes the size of its candidate local stencil.

Let $\theta_{LRBF} = \{(b_R^{(s)}, c_s) : s = 1, \dots, N\}$ denote the set of learnable parameters associated with the Local RBF approximation. Since this work emphasizes adaptively supported domain, the shape parameter c_s is fixed throughout all experiments. The trainable variables therefore consist of θ_{LRBF} and the unknown vector $\mathbf{u} \in \mathbb{R}^N$.

The overall architecture is illustrated in Fig. 2. The input is the set of collocation points $\mathcal{X}$. The Localization Units perform neighborhood identification and local domain construction. A KD-tree is employed to obtain, for each center $\mathbf{x}_s$, a local support set of N_s neighbors. This operation is encoded as a Boolean index matrix $\mathbf{P}^{(s)} \in \{0, 1\}^{N_s \times N}$, with each row containing a single unit element indicating the selected global node. The collection of all local index forms $\{\mathbf{P}^{(s)}\}_{s=1}^N$.

The global pairwise distance matrix $\mathbf{D} \in \mathbb{R}^{N \times N}$ is constructed as $\mathbf{D}_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$, which is symmetric with zero diagonal entries. Localization is achieved by the projection

$$\mathbf{D}^{(s)} = \mathbf{P}^{(s)} \mathbf{D} (\mathbf{P}^{(s)})^T, \quad (23)$$

producing a local distance matrix of size $N_s \times N_s$. The first-row local distance vector is denoted

$$\mathbf{r}^{(s)} = \mathbf{e}_1^T \mathbf{P}^{(s)} \mathbf{D} (\mathbf{P}^{(s)})^T, \quad \mathbf{e}_1 = [1, 0, \dots, 0]^T. \quad (24)$$

Each localized distance matrix $\mathbf{D}^{(s)}$ is processed by the RBF Layer that applies a parameterized radial basis mapping

$$\mathbf{A}^{(s)} = \phi(\mathbf{D}^{(s)}; c_s). \quad (25)$$

A differential RBF layer then applies the operator

$$\mathcal{M}\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_j^{(s)}\|) = \begin{cases} \mathcal{L}\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_j^{(s)}\|), & \mathbf{x}_0^{(s)} \in \mathcal{X}_{\text{int}}, \\ \mathcal{B}\phi(\|\mathbf{x}_0^{(s)} - \mathbf{x}_j^{(s)}\|), & \mathbf{x}_0^{(s)} \in \mathcal{X}_{\text{bd}}, \end{cases} \quad (26)$$

producing the operator-weighted vector $\mathcal{M}\Phi^{(s)} \in \mathbb{R}^{1 \times N_s}$.

The Local Gated Units introduce learnable sparsification within each local matrix $\mathbf{A}^{(s)}$. A gating function $\mathcal{G}_s(\cdot; b_R^{(s)})$ parameterized by a learnable bias $b_R^{(s)}$ determines the effective support domain adaptively. For monotonically decreasing RBFs, a gating mechanism based on a sigmoid mask is employed:

$$\mathcal{G}_s(\mathbf{A}^{(s)}, b_R^{(s)}) = \mathbf{G}^{(s)} \odot \mathbf{A}^{(s)}, \quad \text{with } \mathbf{G}^{(s)} = \sigma(\mathbf{A}^{(s)} - b_R^{(s)}), \quad (27)$$

where $\sigma(\cdot)$ denotes the sigmoid activation function. For monotonically increasing RBFs, the mask is defined as $\mathbf{G}^{(s)} = \sigma(b_R^{(s)} - \mathbf{A}^{(s)})$, ensuring the gate response matches the desired monotonicity.

The output of this unit is

$$\tilde{\mathbf{A}}^{(s)} = \mathcal{G}_s(\mathbf{A}^{(s)}, b_R^{(s)}), \quad (28)$$

which is typically sparser than the original $\mathbf{A}^{(s)}$. The bias parameters $\{b_R^{(s)}\}$ are trainable, allowing the model to learn the optimal local support adaptively.

Following the classical Local RBF collocation mechanism, each local system forms

$$\tilde{\Theta}^{(s)} = \mathcal{M}\Phi^{(s)} (\tilde{\mathbf{A}}^{(s)})^{-1}, \quad (29)$$

yielding one row of the global operator through assembly:

$$\tilde{\Theta}_s = \tilde{\Theta}^{(s)} \mathbf{P}^{(s)}, \quad \tilde{\Theta} = \begin{bmatrix} \tilde{\Theta}_1 \\ \vdots \\ \tilde{\Theta}_N \end{bmatrix}. \quad (30)$$

The final prediction is obtained by the inner product

$$Y_s = v(\tilde{\Theta}_s, \mathbf{u}) = \langle \tilde{\Theta}_s, \mathbf{u} \rangle, \quad (31)$$

which can be interpreted as the output neuron of the s th collocation point.

3.2. Loss formulation and parameter initialization strategy

The reconstruction of the PDEs solution is formulated as the joint minimization problem

$$(\theta_{LRBF}, \mathbf{u}) = \arg \min_{\theta_{LRBF}, \mathbf{u}} \mathcal{J}(\theta_{LRBF}, \mathbf{u}), \quad (32)$$

where the total loss functional $\mathcal{J}$ is defined as

$$\mathcal{J}(\theta_{LRBF}, \mathbf{u}) = \lambda_1 \mathcal{J}_v + \lambda_2 \mathcal{J}_D. \quad (33)$$

The structural residual loss is given by

$$\mathcal{J}_v = \frac{1}{2} \|\tilde{\Theta}(\theta_{LRBF}) \mathbf{u} - \mathbf{g}\|_2^2, \quad (34)$$

which has been demonstrated to be effective in previous studies (Han et al., 2025). The Dirichlet boundary discrepancy is penalized via

$$\mathcal{J}_D = \frac{1}{N_D} \sum_{i=1}^{N_D} |u(\mathbf{x}_i) - u^*(\mathbf{x}_i)|^2. \quad (35)$$

The numerical unknowns $\mathbf{u}$ are initialized via a normal distribution. For the adaptive radii $\{b_R^{(s)}\}_{s=1}^N$, the initialization is based on the numerical characteristics of the local RBF matrices $\{\mathbf{A}^{(s)}\}_{s=1}^N$. All local matrices are stacked into a third-order tensor $\mathcal{A} \in \mathbb{R}^{N_s \times N_s \times N}$, where N_s is the number of nodes in each local stencil. The tensor is vectorized as $\mathbf{a} = \text{vec}(\mathcal{A})$, providing a pool of representative interaction values from all local systems. An empirical distribution $F_{\mathcal{A}}$ is constructed from the entries of $\mathbf{a}$, and the initial radius for each stencil is assigned using the quantile $Q_{\alpha}(F_{\mathcal{A}})$ with $0 < \alpha < 1$. The choice of α ensures that the initial $b_R^{(s)}$ reflects interaction levels near the boundary of the local support domain, leading to a physically meaningful and numerically stable starting point for subsequent training.

The optimization variables $(\theta_{LRBF}, \mathbf{u})$ are updated using gradient-based backpropagation with the Adam optimizer. Distinct learning rates are employed for the adaptive radii and the nodal unknowns, denoted by η_b for the parameters $\{b_R^{(s)}\}$ and η_u for the solution vector $\mathbf{u}$. To enhance

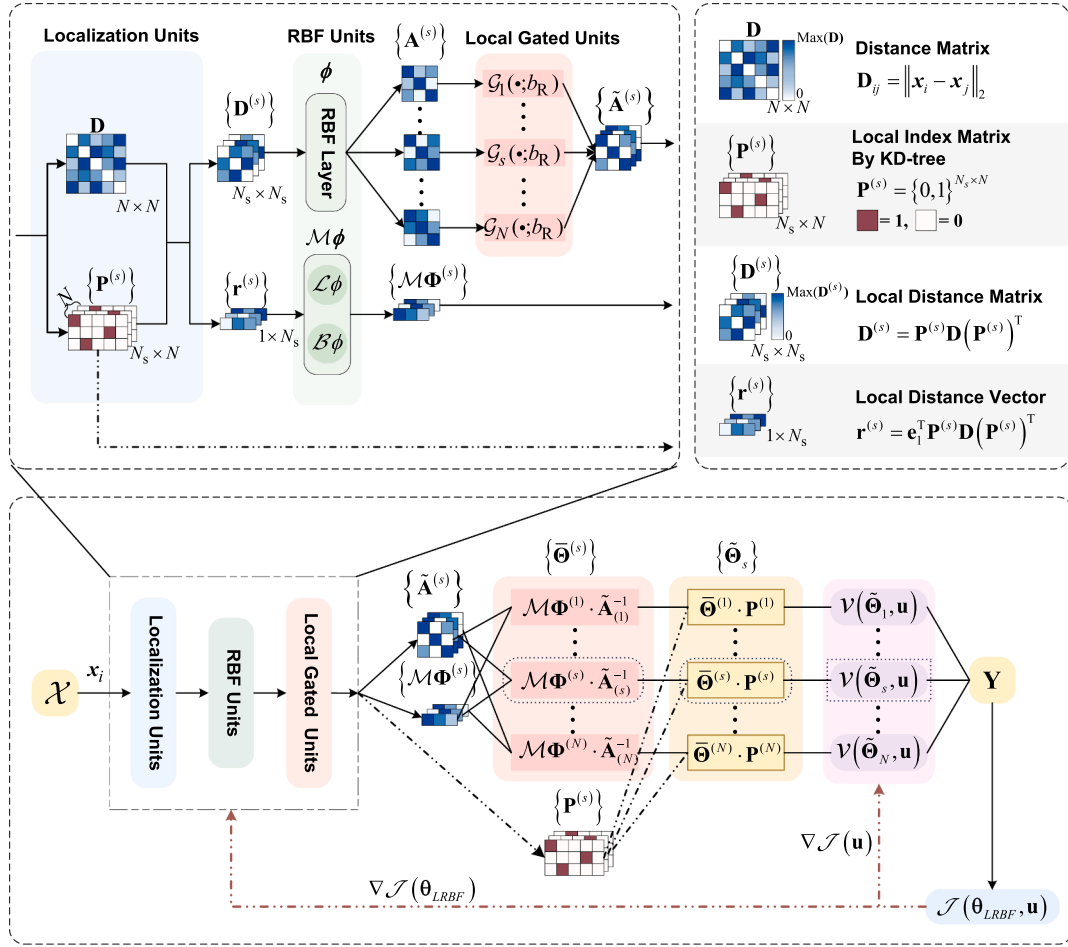


Fig. 2. Network structure of the local RBF-ASD.

convergence stability, a multiplicative learning-rate decay strategy is adopted. Whenever the loss fails to decrease over a prescribed number of epochs, the learning rate is updated according to $\eta^{(r+1)} = \gamma \eta^{(r)}$ where $\gamma < 1$ denotes the decay factor.

It is useful to distinguish the proposed adaptive mechanism from conventional hyperparameter optimization. In a standard tuning procedure, the stencil size is selected through an outer-loop problem of the form

$$N_s^* = \arg \min_{N_s \in \mathcal{S}} \mathcal{E}(N_s), \quad (36)$$

where each candidate N_s requires an independent construction and solution of the Local RBF system, and the resulting value is normally shared by all centers. The criterion $\mathcal{E}$ may be based on an exact-solution error in benchmark problems or on another prescribed indicator.

In Local RBF-ASD, by contrast, N_s defines a candidate stencil, while the center-dependent gate parameters $\{b_R^{(s)}\}_{s=1}^N$ are internal trainable variables. They are determined together with the nodal solution through

$$(\{b_R^{(s)}\}_{s=1}^N, \mathbf{u}) = \arg \min \mathcal{J}. \quad (37)$$

Consequently, support adaptation is performed inside the PDE-solving process rather than through repeated outer-loop evaluations. Since sigmoid gates are used, the learned support should be interpreted as an effective or softly weighted support rather than an exactly discrete stencil cardinality. This formulation allows different centers to retain different effective interactions, while the candidate value N_s serves only as an upper bound on the available local neighbors.

This mechanism improves numerical robustness and promotes stable convergence of the adaptive RBF architecture. The resulting Local RBF-ASD framework may therefore be interpreted as a differentiable optimization model in which meshfree approximation and learnable support-domain adaptation are coupled within a unified mathematical formulation.

3.3. Model complexity and computational cost

For a fixed candidate stencil size N_s , Local RBF and Local RBF-ASD use the same candidate neighboring nodes and local matrices of the same dimensions. The comparison under the same N_s therefore isolates the effect of the adaptive gating mechanism, although the two methods do not contain the same number of optimization variables.

Let N_u denote the number of nodal unknowns and let N denote the number of collocation centers. The numbers of unknown or optimization variables are $P_{\text{Local RBF}} = N_u$, $P_{\text{Local RBF-ASD}} = N_u + N$.

For the scalar PDE considered here, $N_u = N$, and hence Local RBF and Local RBF-ASD contain N and $2N$ variables, respectively. The additional N variables correspond to the center-dependent gating parameters $\{b_R^{(s)}\}_{s=1}^N$.

Both methods store local matrices of size $N_s \times N_s$ for the N collocation centers. Their leading matrix-storage requirement is therefore of order $\mathcal{O}(NN_s^2)$. Local RBF-ASD additionally requires $\mathcal{O}(N)$ storage for the gating parameters. Although this does not change the leading asymptotic storage order, the practical memory consumption of Local RBF-ASD is higher because gradient information, optimizer states, and

intermediate tensors required by automatic differentiation must also be retained.

Local RBF-ASD is not intended to reduce the computational time of a single PDE solve. Its main objective is to reduce the sensitivity of the numerical solution to the prescribed candidate stencil size N_s .

4. Numerical results

In this section, several numerical experiments are conducted to evaluate the performance of the proposed Local RBF-ASD method on a series of benchmark problems. The accuracy of the numerical solutions is assessed using the pointwise maximum error and the relative L^2 error, defined as

$$\begin{aligned}\epsilon_{\max} &:= \max_{i=1,\dots,N} |u(\mathbf{x}_i) - u^*(\mathbf{x}_i)|, \\ \epsilon_{\text{rel } L^2} &:= \frac{\left(\sum_{i=1}^N |u(\mathbf{x}_i) - u^*(\mathbf{x}_i)|^2\right)^{1/2}}{\left(\sum_{i=1}^N |u^*(\mathbf{x}_i)|^2\right)^{1/2}},\end{aligned}\quad (38)$$

where $\{\mathbf{x}_i\}_{i=1}^N$ denotes a set of evaluation points uniformly sampled in the computational domain, $u(\mathbf{x}_i)$ is the numerical approximation, and $u^*(\mathbf{x}_i)$ is the exact solution. The maximum error $\epsilon_{\max}$ is unnormalized, whereas the relative L^2 error $\epsilon_{\text{rel } L^2}$ is normalized by the discrete L^2 norm of the exact solution.

All computations are performed on a Windows 11 system equipped with an Intel® Core™ i9-185H CPU (2.50–5.10 GHz), 32 GB of RAM, and an NVIDIA® GeForce RTX 4060 Laptop GPU with 8 GB of dedicated memory. The approach are implemented using the PyTorch library (version 2.5.1).

4.1. 2D Helmholtz equation

To evaluate the stability and accuracy of the proposed learning-based local support domain adaptation method, we investigate the numerical performance on a two-dimensional Helmholtz equation defined over an irregular, amoeba-shaped domain Ω . The governing equation is expressed as:

$$\nabla^2 u(x, y) + \kappa^2 u(x, y) = 0, \quad (x, y) \in \Omega, \quad (39)$$

$$Bu(x, y) = h(x, y), \quad (x, y) \in \partial\Omega, \quad (40)$$

where κ denotes the wave number, where $h(x, y)$ of related Dirichlet and Neumann boundary conditions can be derived from exact solutions $u^*(x, y) = \sin(\kappa x) + \cos(\kappa y)$. The boundary of the computational domain Ω is described by the parametric function:

$$\partial\Omega = \{(x, y) : x = r(\theta) \cos(\theta), y = r(\theta) \sin(\theta), 0 \leq \theta \leq 2\pi\}, \quad (41)$$

with radial function

$$r(\theta) = \rho(e^{\sin(\theta)} \sin^2(2\theta) + e^{\cos(\theta)} \cos^2(2\theta)), \quad (42)$$

where $\rho = 0.5$ controls the size of the domain. This function generates a nonconvex amoeba-like geometry, as illustrated in Fig. 3.

To assess the performance of the Local RBF and Local RBF-ASD methods, we adopt a common numerical setup in which collocation points are uniformly sampled on a 70×70 grid, yielding $N_{\text{int}} = 2448$ interior points and $N_{\text{bd}} = 500$ boundary points, and Gaussian RBFs are employed as basis functions. To investigate the influence of the number of local nodes N_s , a series of experiments is carried out with fixed $\kappa = 1$ under several choices of the shape parameter c_s . As shown in Fig. 4, the pointwise maximum error is recorded for N_s ranging from 35 to 60 for both methods with $c_s \in \{3, 5, 7\}$.

Numerical errors of the Local RBF-ASD method in Fig. 4a fluctuate between 10^{-2} and 10^{-4} , suggesting stable performance. In contrast, the Local RBF approach exhibits larger variations, with errors ranging from about 10^{-1} down to 10^{-5} . Although the Local RBF method achieves

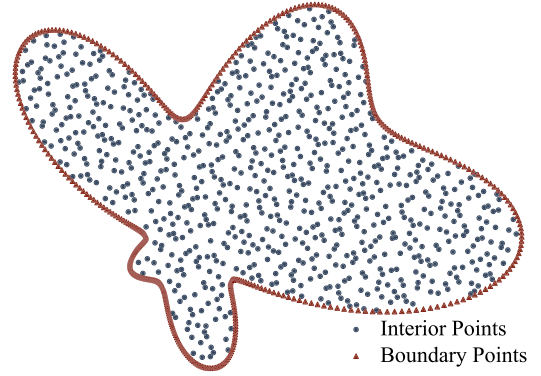


Fig. 3. Amoeba-shaped computational domain Ω (Example 4.1).

higher accuracy when N_s is large, its performance is less consistent, occasionally dropping to an accuracy level of only 10^{-1} . Therefore, the Local RBF-ASD method demonstrates greater stability, providing a more reliable level of accuracy overall. This reflects a typical trade-off between accuracy and stability. For $c_s = 5$ and $c_s = 7$, as shown in Fig. 4b and c, the Local RBF-ASD method maintains errors around 10^{-3} , while the Local RBF method exhibits significantly larger fluctuations reaching up to 10^{-1} for some N_s values. In most cases, the Local RBF-ASD method attains higher accuracy than the Local RBF method. Overall, Local RBF-ASD demonstrates markedly better stability and more consistent performance.

As shown in Fig. 5a, the loss function exhibits a clear and consistent decreasing trend throughout the Local RBF-ASD training process. The training process consists of two distinct phases. In the initial phase (before approximately 150 epochs), due to the relatively large learning rate for b_R , the training primarily explores the parameter space for the optimal b_R , resulting in fluctuating loss values. After learning rate adjustment through the scheduling strategy, the process enters the second phase, where it focuses on learning in the solution space, and the loss function shows a continuous decreasing trend. The loss stabilizes around 300 epochs, indicating that the model has successfully minimized the loss to achieve the training objective. The total loss $\mathcal{J}$ and structural loss $\mathcal{J}_v$ follow similar trends, demonstrating that the optimization process is primarily constrained by $\mathcal{J}_v$. The consistent evolution of the pointwise maximum error $\epsilon_{\max}$ with the total loss $\mathcal{J}$ indicates that $\mathcal{J}$ effectively guides the learning and optimization process of the solution u . As illustrated in Fig. 5b, the relative L^2 error decreases consistently with the reduction of the training loss. The data points exhibit an approximately linear trend in logarithmic scales, with a fitted slope close to $1/2$. This indicates that the relative L^2 error and the training loss roughly follow a square-root relation, showing that the loss provides a reliable measure of the solution accuracy.

To evaluate the stability of Local RBF and Local RBF-ASD with respect to N_s under different wave numbers κ , we observe from Table 2 that the proposed method maintains stable accuracy as the number of local support nodes N_s varies. The $\epsilon_{\max}$ and $\epsilon_{\text{rel } L^2}$ of Local RBF-ASD are generally on the order of 10^{-3} and 10^{-4} , respectively, while Local RBF exhibits significant error fluctuations with changes in N_s . Particularly when $N_s = 35$, the error of Local RBF increases substantially. The Local RBF-ASD method maintains low pointwise maximum errors throughout the tested range for different wave numbers κ . This indicates that the method is numerically robust and insensitive to the specific choice of N_s , effectively avoiding the ill-conditioning issues that frequently arise in dense local interpolation systems. Specifically, the Local RBF-ASD method achieves smaller and more uniform errors, confirming that the adaptive directional strategy further enhances the stability and reliability of local approximations.

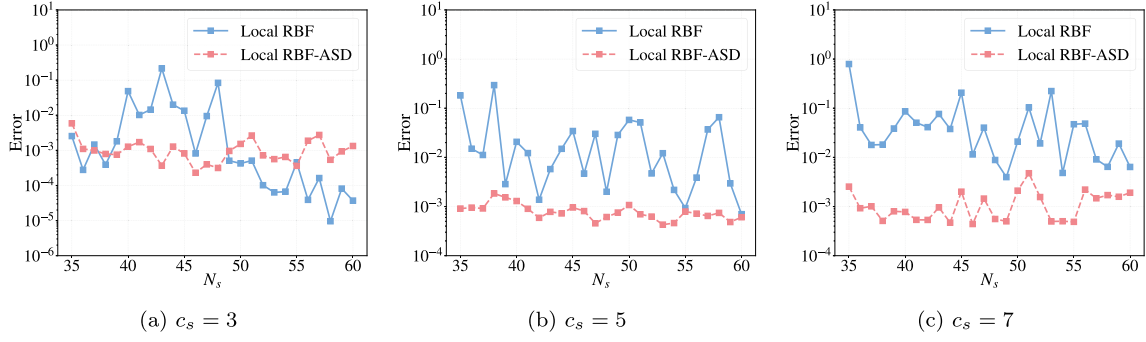


Fig. 4. Variation of the error $\varepsilon_{\max}$ with respect to the number of local support nodes N_s under different shape parameters c_s (Example 4.1).

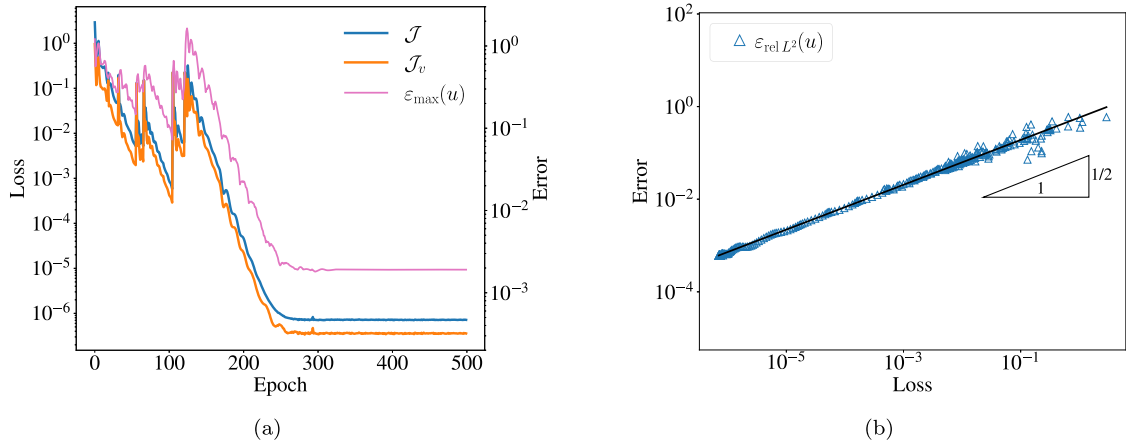


Fig. 5. Training loss and error evolution for the Local RBF-ASD method in Example 4.1 with $\kappa = 1$, $N_s = 45$ and $c_s = 5$: (a) evolution of $\mathcal{J}$, $\mathcal{J}_v$, and $\varepsilon_{\max}$ with epochs; (b) relation between the relative L^2 error $\varepsilon_{\text{rel } L^2}$ and the total loss $\mathcal{J}$.

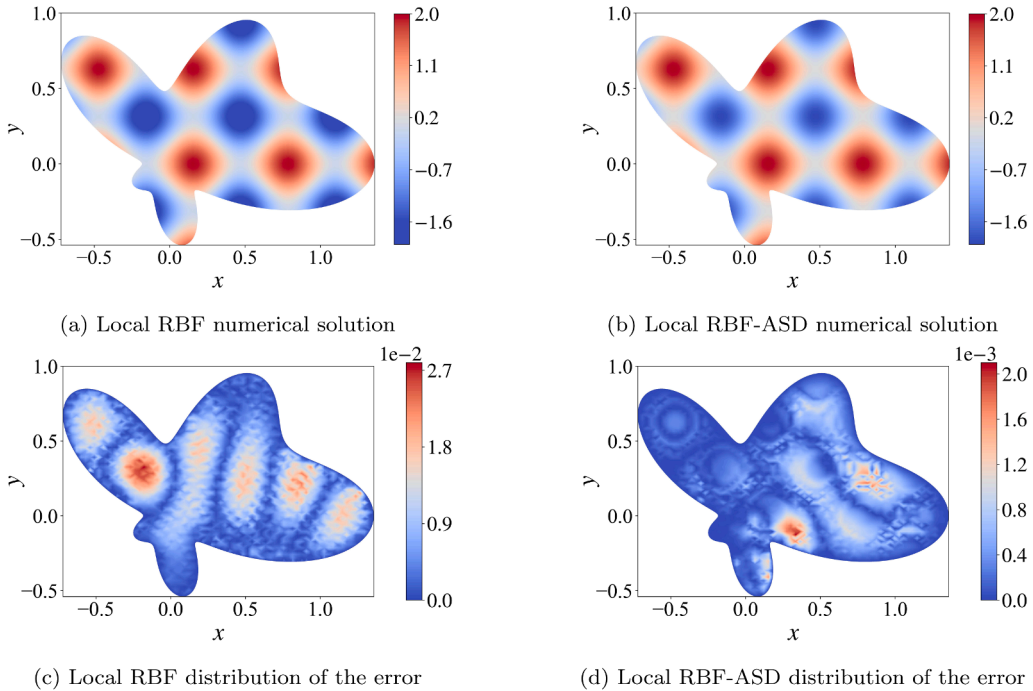


Fig. 6. Numerical results with $\kappa = 10$, $N_s = 45$ and $c_s = 5$ (Example 4.1).

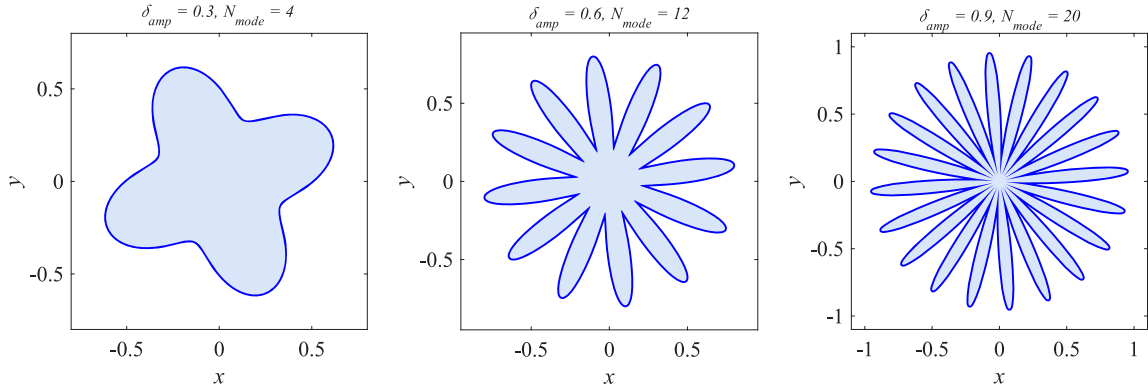


Fig. 7. Flower-shaped domains generated using different values of (δ_{amp}, N_{mode}) .

Table 2

Error stability across local support sizes N_s under different wave numbers κ with $c_s = 5$ (Example 4.1).

κ	N_s	Local RBF		Local RBF-ASD	
		$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$	$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$
1	35	1.8202×10^{-1}	1.5527×10^{-2}	9.0889×10^{-4}	3.2845×10^{-4}
1	40	2.0725×10^{-2}	1.3956×10^{-3}	1.2946×10^{-3}	1.5530×10^{-4}
1	45	3.4375×10^{-2}	3.0148×10^{-3}	9.5682×10^{-4}	2.6096×10^{-4}
1	50	5.7924×10^{-2}	3.2494×10^{-3}	1.0731×10^{-3}	1.1041×10^{-4}
1	55	9.3094×10^{-4}	5.4958×10^{-5}	7.9176×10^{-4}	1.6111×10^{-4}
1	60	6.9899×10^{-4}	1.5554×10^{-4}	6.0625×10^{-4}	1.0203×10^{-4}
5	35	1.1181×10^0	1.4878×10^{-1}	1.1073×10^{-3}	3.9886×10^{-4}
5	40	5.2166×10^{-2}	4.3992×10^{-3}	6.5809×10^{-4}	1.8620×10^{-4}
5	45	7.8798×10^{-3}	6.8960×10^{-4}	5.2422×10^{-4}	1.7578×10^{-4}
5	50	2.4700×10^{-2}	2.0785×10^{-3}	8.1007×10^{-4}	1.9633×10^{-4}
5	55	2.3306×10^{-3}	4.1002×10^{-4}	1.2778×10^{-3}	3.5640×10^{-4}
5	60	3.5317×10^{-4}	6.3259×10^{-5}	6.0936×10^{-4}	1.6382×10^{-4}
10	35	5.5964×10^0	7.7538×10^{-1}	9.1230×10^{-3}	2.1724×10^{-3}
10	40	1.9788×10^{-3}	5.6470×10^{-4}	5.9543×10^{-3}	1.5424×10^{-3}
10	45	2.7806×10^{-2}	8.5341×10^{-3}	2.0857×10^{-3}	4.3489×10^{-4}
10	50	3.7507×10^{-3}	6.5590×10^{-4}	2.5634×10^{-3}	9.9016×10^{-4}
10	55	4.9355×10^{-4}	9.5036×10^{-5}	8.9064×10^{-4}	1.8610×10^{-4}
10	60	1.6998×10^{-3}	2.7412×10^{-4}	1.2492×10^{-3}	3.2906×10^{-4}

Furthermore, Fig. 6 presents the numerical results for the high wave number case $\kappa = 10$ with $N_s = 45$ and $c_s = 5$. When the Local RBF method fails to produce stable solutions, resulting in a pointwise maximum error of 2.7×10^{-2} , the Local RBF-ASD method achieves significantly improved results with error range of 0 to 2×10^{-3} , demonstrating effective and stable learning for solving the equation.

As shown in Table 3, Local RBF-ASD without learning-rate decay still produces substantially smaller errors than the conventional Local RBF method for all three wave numbers. This indicates that the main improvement originates from the adaptive support-domain mechanism rather than from the learning-rate schedule.

Learning-rate decay further reduces the relative L^2 error in all three cases and reduces the maximum error for $\kappa = 5$ and $\kappa = 10$. However, it does not improve every error measure, as the maximum error for $\kappa = 1$ is smaller without decay. Therefore, learning-rate decay is regarded as an auxiliary optimization strategy rather than the source of the improved performance.

$$r(\theta) = \frac{1}{2}(1 + \delta \sin(N\theta)), \quad \theta \in [0, 2\pi], \quad (43)$$

The computational domain Ω is enclosed by the boundary defined in polar coordinates by (43). Here, δ_{amp} controls the amplitude of the boundary perturbation, while N_{mode} determines the number of oscillations, or petals, along the boundary. Therefore, varying (δ_{amp}, N_{mode})

produces solution domains with different geometric complexities, as illustrated in Fig. 7.

To investigate the influence of the solution-domain geometry, we further evaluate the performance of Local RBF and Local RBF-ASD on three flower-shaped domains characterized by distinct parameter pairs (δ_{amp}, N_{mode}) . A consistent numerical setup is adopted, wherein collocation points are uniformly sampled on a 70×70 grid, resulting in $N_{\text{int}} = 2468$ interior points and $N_{\text{bd}} = 1200$ boundary points, with Gaussian RBFs employed as the basis functions. The remaining numerical parameters are fixed at $N_s = 45$, $c_s = 5$, and $\kappa = 10$ to ensure that the effects of geometric variation can be assessed in a controlled manner.

As shown in Table 4, Local RBF-ASD consistently produces smaller relative L^2 errors and generally smaller maximum errors across the three geometrically different domains, demonstrating its robustness with respect to variations in the solution-domain shape.

4.2. 2D anisotropic Helmholtz equation

The physical problem concerns steady-state heat conduction in a two-dimensional anisotropic and spatially graded medium Ω with no internal sources (Marin & Lesnic, 2007). The governing equation takes the form:

$$\sum_{i,j=1}^2 \frac{\partial}{\partial x_i} \left(\Lambda_{ij}(\mathbf{x}, T) \frac{\partial T(\mathbf{x})}{\partial x_j} \right) = 0, \quad \mathbf{x} \in \Omega, \quad (44)$$

where $T(\mathbf{x})$ is the temperature and $\Lambda(\mathbf{x}, T) = [\Lambda_{ij}(\mathbf{x}, T)]$ is a symmetric positive-definite conductivity tensor ($\Lambda_{12} = \Lambda_{21}$, $\det(\Lambda) > 0$). We assume an exponentially graded medium

$$\Lambda_{ij}(\mathbf{x}, T) = \alpha(T) \tilde{\Lambda}_{ij} e^{2\beta \cdot \mathbf{x}}, \quad \mathbf{x} \in \Omega, \quad (45)$$

where $\alpha(T) > 0$, $\tilde{\Lambda} = [\tilde{\Lambda}_{ij}]$ is a constant symmetric matrix with $\det(\tilde{\Lambda}) > 0$, and $\beta = (\beta_1, \beta_2)^T$ is a constant grading vector.

Introduce the Kirchhoff transformation

$$\psi(T) = \int \alpha(T) dT, \quad \Psi(\mathbf{x}) = \psi(T(\mathbf{x})), \quad (46)$$

which is monotone and invertible since $\alpha(T) > 0$. Substitution of Eq. (46) into Eq. (44) gives

$$\sum_{i,j=1}^2 \tilde{\Lambda}_{ij} \Psi_{x_i x_j} + 2\beta^T (\tilde{\Lambda} \nabla \Psi) = 0, \quad \mathbf{x} \in \Omega. \quad (47)$$

To eliminate the first-order term, define the auxiliary field

$$u(\mathbf{x}) = \Psi(\mathbf{x}) e^{\beta \cdot \mathbf{x}}, \quad (48)$$

which yields the modified Helmholtz equation

$$\sum_{i,j=1}^2 \tilde{\Lambda}_{ij} u_{x_i x_j} - (\beta^T \tilde{\Lambda} \beta) u = 0, \quad \mathbf{x} \in \Omega, \quad (49)$$

Table 3Ablation study of learning-rate decay with $N_s = 45$ and $c_s = 5$ (Example 4.1).

κ	Local RBF		Local RBF-ASD w/o decay		Local RBF-ASD w/ decay	
	$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$	$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$	$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$
1	3.4375×10^{-2}	3.0148×10^{-3}	4.4683×10^{-4}	1.3321×10^{-4}	9.5682×10^{-4}	9.8645×10^{-5}
5	7.8798×10^{-3}	6.8960×10^{-4}	1.7293×10^{-3}	2.5765×10^{-4}	5.2422×10^{-4}	1.7578×10^{-4}
10	2.7806×10^{-2}	8.5341×10^{-3}	2.9625×10^{-3}	9.5285×10^{-4}	2.0857×10^{-3}	4.3489×10^{-4}

Table 4Error comparison for the different flower-shaped domain with $N_s = 45$, $c_s = 5$ and $\kappa = 10$ (Example 4.1).

$(\delta_{\text{amp}}, N_{\text{mode}})$	Local RBF		Local RBF-ASD	
	$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$	$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$
(0.3, 4)	2.6616×10^{-3}	7.5296×10^{-4}	5.6135×10^{-5}	1.5448×10^{-5}
(0.6, 12)	7.3026×10^{-4}	9.0334×10^{-5}	3.9329×10^{-5}	4.5451×10^{-6}
(0.9, 20)	1.6854×10^{-3}	9.3802×10^{-5}	2.0507×10^{-3}	6.1136×10^{-5}

with $\kappa^2 = \beta^T \tilde{\Lambda} \beta > 0$.

For an orthotropic case,

$$\tilde{\Lambda}_{12} = \tilde{\Lambda}_{21} = 0, \quad \tilde{\Lambda}_{11} = 2, \quad \tilde{\Lambda}_{22} = 1, \quad (50)$$

and $\beta = (0, 1)^T$, we have $\kappa^2 = 1$. Consider a linear change of variables $\xi = Qx$ with $Q \tilde{\Lambda} Q^T = I$, e.g., $Q = \text{diag}(1/\sqrt{2}, 1)$ so that $\xi_1 = x_1/\sqrt{2}$ and $\xi_2 = x_2$. Then Eq. (49) reduces to the canonical form, which is analogous to Eq. (40):

$$\Delta_\xi u(\xi) - \kappa^2 u(\xi) = 0, \quad \xi \in Q(\Omega). \quad (51)$$

For $\kappa^2 = 1$, a closed-form solution is

$$u^*(\xi) = \frac{\sinh(\rho)}{\sqrt{2}\rho} \sqrt{1 + \frac{\xi_1 - 1}{\rho}}, \quad (52)$$

where $\rho = \sqrt{(\xi_1 - 1)^2 + \xi_2^2}$. Transforming back via Eq. (48),

$$\Psi^*(x) = \frac{\sqrt{1 - \chi/\rho}}{\sqrt{2}\rho} \sinh(\rho) e^{-x_2}, \quad (53)$$

where $\chi = \frac{x_1}{\sqrt{2}} - 1$, $\rho = \sqrt{\chi^2 + x_2^2}$. The analytical temperature field follows from the inverse Kirchhoff transform $T = \psi^{-1}(\Psi)$. Following the form in Sawaf et al. (1995), the thermal conductivity is modeled as linearly temperature-dependent: $\alpha(T) = 1 + \gamma T$ with $\gamma > 0$,

$$\psi(T) = T + \frac{\gamma}{2} T^2, \quad T = \frac{-1 + \sqrt{1 + 2\gamma\Psi}}{\gamma}, \quad (54)$$

so that, when $\gamma = \frac{1}{2}$ one obtains the compact form

$$T^*(x) = 2\sqrt{\Psi^*(x) + 1} - 2, \quad x \in \Omega. \quad (55)$$

Dirichlet boundary conditions are imposed on $\partial\Omega$,

$$T(x) = f(x), \quad x \in \partial\Omega, \quad (56)$$

and, by Eq. (46), the transformed boundary conditions for Ψ remain Dirichlet. For $\alpha(T) = 1 + \gamma T$ with $\gamma = \frac{1}{2}$,

$$\Psi(x) = f(x) + \frac{1}{4} f^2(x), \quad x \in \partial\Omega. \quad (57)$$

Consider Eq. (44) on the square domain $\Omega = (0, 1) \times (0, 1)$, where the system given by Eqs. (47), (57) is equivalent to the original nonlinear conduction problem Eqs. (44), (56).

To further demonstrate the stability of the Local RBF-ASD method for solving the heterogeneous Helmholtz equation under the adaptively supported domain mechanism, a test case is considered in which a uniform grid of 40×40 collocation points is distributed inside the computational domain, and 500 boundary points are uniformly placed along $\partial\Omega$. The Gaussian RBFs are employed as basis functions.

Table 5Error stability across local support sizes N_s for different RBF types with $c_s = 5$ (Example 4.2).

N	N_s	Local RBF		Local RBF-ASD	
		$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$	$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$
1084	40	4.0393×10^{-4}	2.5100×10^{-4}	8.1028×10^{-5}	7.7256×10^{-5}
1084	50	1.3532×10^{-4}	1.0436×10^{-4}	1.0611×10^{-4}	1.0714×10^{-4}
1084	60	1.5571×10^{-4}	1.5256×10^{-4}	1.2663×10^{-4}	1.1229×10^{-4}
1944	40	7.7342×10^{-4}	3.3577×10^{-4}	1.7896×10^{-5}	1.7594×10^{-5}
1944	50	8.3740×10^{-5}	2.7490×10^{-5}	5.3790×10^{-5}	2.5553×10^{-5}
1944	60	4.7104×10^{-4}	1.3076×10^{-4}	2.2422×10^{-5}	2.4559×10^{-5}
2804	40	9.6092×10^{-5}	5.5184×10^{-5}	1.3530×10^{-5}	6.2915×10^{-6}
2804	50	6.6201×10^{-5}	2.7847×10^{-5}	4.4008×10^{-5}	1.7433×10^{-5}
2804	60	6.5693×10^{-3}	3.5290×10^{-3}	7.6032×10^{-5}	4.0827×10^{-5}
3764	40	1.9908×10^{-3}	7.9165×10^{-4}	6.3618×10^{-5}	4.6651×10^{-5}
3764	50	7.9785×10^{-6}	5.9036×10^{-6}	3.0028×10^{-5}	1.5866×10^{-6}
3764	60	1.6696×10^{-4}	7.9140×10^{-5}	2.1727×10^{-5}	1.4011×10^{-5}
5224	40	2.1095×10^{-3}	5.9930×10^{-4}	6.1253×10^{-5}	3.4339×10^{-5}
5224	50	1.4110×10^{-4}	6.4771×10^{-5}	1.9747×10^{-5}	1.1494×10^{-5}
5224	60	1.3973×10^{-4}	7.4119×10^{-5}	2.7392×10^{-5}	1.0148×10^{-5}

Table 6Error comparison of Local RBF and Local RBF-ASD for different RBF types and local support sizes N_s (Example 4.2).

RBFs	N_s	Local RBF		Local RBF-ASD	
		$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$	$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$
MQ	40	2.2125×10^{-4}	1.4643×10^{-4}	7.0850×10^{-5}	5.9492×10^{-5}
	45	9.3626×10^{-5}	7.3688×10^{-5}	5.2511×10^{-5}	4.3388×10^{-5}
	50	1.7524×10^{-4}	1.5192×10^{-4}	5.4338×10^{-5}	4.4861×10^{-5}
	55	1.7429×10^{-4}	1.4308×10^{-4}	5.2346×10^{-5}	4.2025×10^{-5}
	60	2.6804×10^{-5}	2.2084×10^{-5}	5.2241×10^{-5}	3.7758×10^{-5}
IMQ	40	7.5324×10^{-4}	6.3751×10^{-4}	3.5686×10^{-4}	3.2619×10^{-4}
	45	5.0252×10^{-4}	3.9816×10^{-4}	2.0170×10^{-4}	1.6657×10^{-4}
	50	1.0164×10^{-3}	8.3735×10^{-4}	1.9216×10^{-4}	1.7452×10^{-4}
	55	1.0423×10^{-3}	8.7757×10^{-4}	2.5510×10^{-4}	2.3164×10^{-4}
	60	1.0853×10^{-4}	9.8522×10^{-5}	2.4902×10^{-4}	2.2981×10^{-4}

A comparison of pointwise maximum errors presented in Fig. 8 reveals a clear contrast between the Local RBF-ASD and Local RBF methods as the number of local nodes N_s increases. The Local RBF-ASD method exhibits consistently smooth error variations, for example between 1×10^{-3} and 1×10^{-5} for the shape parameter $c_s = 3$ in Fig. 8a, whereas the Local RBF method displays a considerably broader and more irregular error profile ranging from 1×10^{-2} to 1×10^{-6} , with distinct sensitivity peaks. This stable-versus-sensitive performance pattern persists for other shape parameters $c_s = 5$ and $c_s = 7$, as shown in Fig. 8b and c. Overall, the proposed Local RBF-ASD approach consistently achieves higher numerical stability and accuracy across a wide range of parameter configurations.

As shown in Fig. 9, the training history of the Local RBF-ASD method for the two-dimensional anisotropic Helmholtz problem illustrates the evolution of both the loss function and the corresponding error metrics with respect to the number of epochs. As observed, the total loss J and the structural loss term J_v decay smoothly and consistently, ultimately reaching magnitudes on the order of 10^{-9} . The training stabilizes

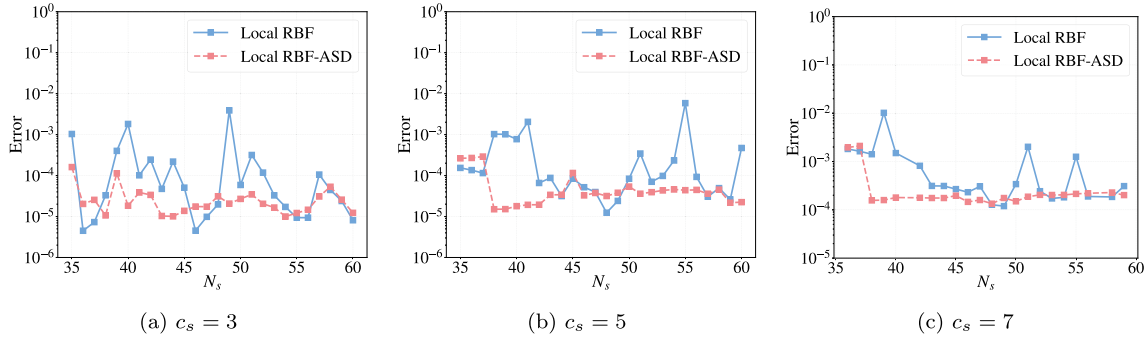


Fig. 8. Variation of the error $\epsilon_{\max}$ with respect to the number of local support nodes N_s under different shape parameters c_s (Example 4.2).

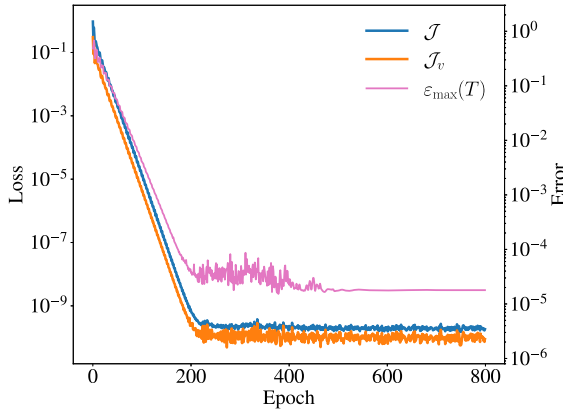


Fig. 9. Loss curve evolution during training with $N_s = 40$ and $c_s = 5$ (Example 4.2).

after approximately 400 epochs, suggesting that the learnable parameters have converged. Correspondingly, the solution errors also decrease and plateau, confirming the consistency between loss minimization and solution accuracy.

To further evaluate the accuracy and stability of the proposed method, a series of tests were conducted using different numbers of collocation points. The total number of interior sampling points, denoted by N , was determined from uniform grids of sizes 30×30 , 40×40 , 50×50 , 60×60 , and 70×70 distributed within the computational domain Ω . The boundary sampling points N_{bd} were selected from predefined sets of $\{300, 400, 500, 600\}$, ensuring sufficient resolution along $\partial\Omega$ for all test configurations. In each case, the same shape parameter $c_s = 5$ was adopted to maintain consistency in the comparison. The detailed numerical errors corresponding to various combinations of N and the local support domain size N_s are summarized in Table 5.

Based on the results in Table 5, the Local RBF-ASD method consistently achieves higher accuracy across all tested configurations. Both the pointwise maximum error $\epsilon_{\max}$ and the relative L^2 error $\epsilon_{\text{rel } L^2}$ are typically at least one order of magnitude smaller than those of the standard Local RBF method. For most cases, the maximum error of the Local RBF-ASD approach remains close to 10^{-5} , while the conventional Local RBF method exhibits larger deviations and irregular fluctuations as the problem size increases.

When the number of interior collocation points N becomes larger, the associated linear system tends to be more ill-conditioned, which often leads to numerical instability in the standard Local RBF formulation. In contrast, the adaptively supported domain mechanism incorporated in the Local RBF-ASD method maintains stable convergence and effectively suppresses error oscillations under the same parameter settings. The relative L^2 error of the proposed method remains consistently low even for the $N = 5224$, demonstrating its robustness with

Table 7

Error stability across local support sizes N_s under different shape parameters c_s (Example 4.3).

c_s	N_s	Local RBF		Local RBF-ASD	
		$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$	$\epsilon_{\max}$	$\epsilon_{\text{rel } L^2}$
3	25	1.8409×10^{-2}	8.8227×10^{-3}	5.3295×10^{-2}	1.7268×10^{-2}
	30	2.0338×10^0	9.1285×10^{-1}	1.5310×10^{-3}	7.0851×10^{-4}
	35	1.1342×10^{-1}	6.0277×10^{-2}	5.9924×10^{-4}	3.7322×10^{-4}
	40	2.4832×10^{-1}	1.0637×10^{-1}	1.2316×10^{-2}	3.1874×10^{-3}
	45	4.0077×10^{-2}	2.0131×10^{-2}	8.7950×10^{-3}	4.6763×10^{-3}
	50	5.2120×10^{-2}	3.4089×10^{-2}	6.2875×10^{-3}	1.9993×10^{-3}
	55	5.9441×10^{-2}	1.7941×10^{-2}	8.6990×10^{-3}	3.6798×10^{-3}
	60	1.3256×10^{-1}	9.1183×10^{-2}	2.7974×10^{-3}	1.1300×10^{-3}
5	25	3.5594×10^{-2}	1.4293×10^{-2}	1.3843×10^{-2}	4.2975×10^{-3}
	30	6.0168×10^{-1}	3.0129×10^{-1}	2.3525×10^{-3}	1.5341×10^{-3}
	35	6.1351×10^{-3}	1.8978×10^{-3}	9.1322×10^{-4}	4.2409×10^{-4}
	40	1.8536×10^{-1}	6.1703×10^{-2}	5.8861×10^{-3}	3.2265×10^{-3}
	45	7.9117×10^{-2}	4.0940×10^{-2}	3.8597×10^{-3}	3.0031×10^{-3}
	50	3.5254×10^{-1}	1.3927×10^{-1}	1.4432×10^{-3}	9.6727×10^{-4}
	55	4.7267×10^{-1}	2.1856×10^{-1}	2.8289×10^{-3}	1.4648×10^{-3}
	60	5.2773×10^{-2}	2.8750×10^{-2}	1.7128×10^{-3}	6.9719×10^{-4}
7	25	4.8860×10^{-1}	2.3751×10^{-1}	1.1117×10^{-2}	3.3437×10^{-3}
	30	7.2939×10^{-2}	3.4728×10^{-2}	3.1862×10^{-3}	1.2346×10^{-3}
	35	2.0012×10^{-3}	9.6309×10^{-4}	9.8639×10^{-4}	5.2209×10^{-4}
	40	3.2019×10^{-1}	1.5891×10^{-1}	2.4281×10^{-3}	1.7367×10^{-3}
	45	1.9932×10^{-1}	8.2862×10^{-2}	6.5149×10^{-3}	2.0443×10^{-3}
	50	4.2486×10^{-2}	1.4837×10^{-2}	2.9959×10^{-3}	1.1970×10^{-3}
	55	3.0989×10^{-2}	1.0965×10^{-2}	2.3697×10^{-3}	1.0154×10^{-3}
	60	4.0995×10^{-2}	1.5666×10^{-2}	3.7319×10^{-3}	3.0815×10^{-3}

Table 8

CPU runtime comparison with $c = 4$.

N_s	Local RBF time (s)	Local RBF-ASD time (s)	Runtime ratio
20	0.2300	168.72	733.57
25	0.2440	225.45	923.98
30	0.2630	279.02	1060.91
35	0.2830	307.35	1086.04
40	0.5305	400.90	755.70
45	1.0300	483.22	469.15
50	1.1010	388.75	353.09
55	1.1950	424.34	355.10
60	1.3080	560.43	428.46

respect to both spatial discretization and problem scale. These observations confirm that the adaptive strategy embedded in the Local RBF-ASD approach significantly enhances its overall reliability for solving anisotropic Helmholtz problems.

Fig. 10 presents the numerical results for the two-dimensional Anisotropic Helmholtz equation with $N = 1944$ and $N_s = 40$. From the morphology of the computed solutions, both the Local RBF and Local RBF-ASD methods accurately capture the essential features of the

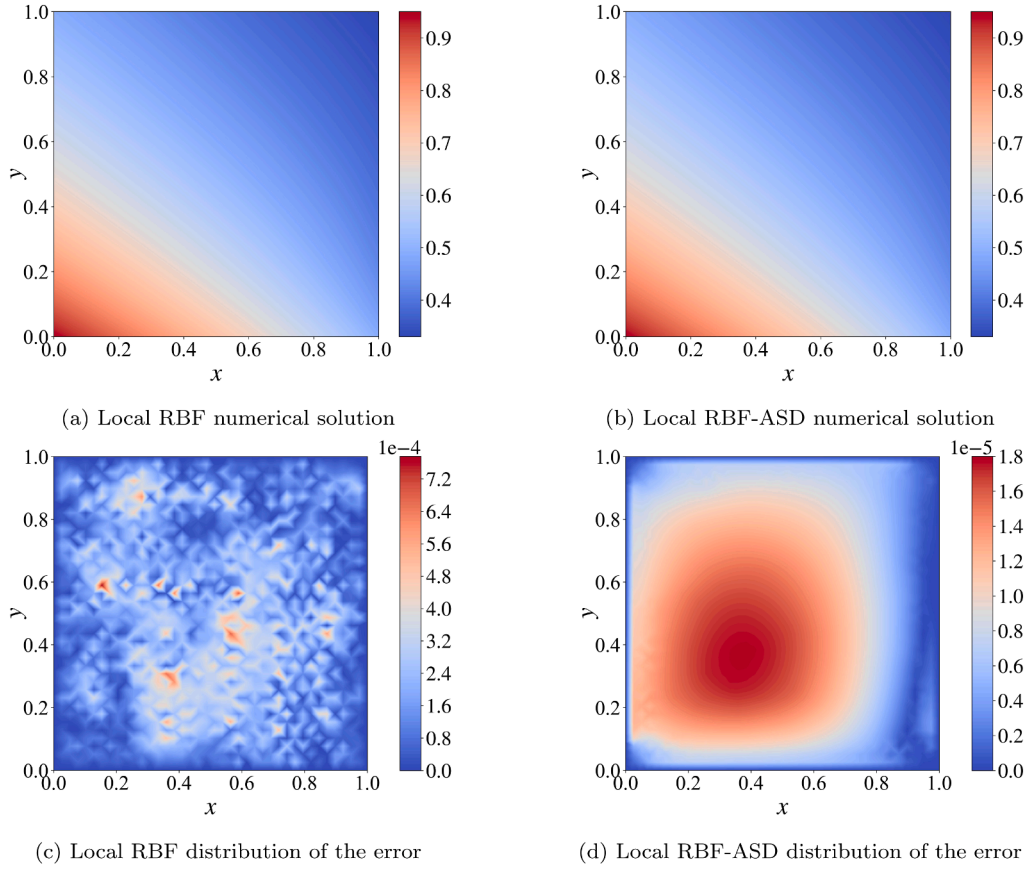


Fig. 10. Numerical results with $N = 1944$, $N_s = 40$ and $c_s = 5$ (Example 4.2).

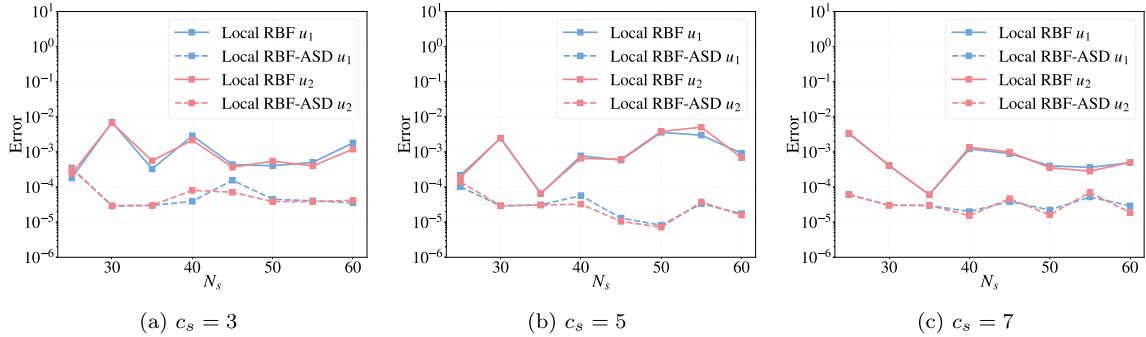


Fig. 11. Variations of errors $\epsilon_{\max}(u_1)$ and $\epsilon_{\max}(u_2)$ with respect to the number of local support nodes N_s for different shape parameters c_s (Example 4.3).

anisotropic field, exhibiting smooth and continuous spatial distributions across the computational domain.

A detailed comparison of the error distributions shown in Fig. 10a and b indicates that the standard Local RBF method exhibits slight instability in certain regions, leading to a noticeable reduction in overall accuracy. In contrast, the Local RBF-ASD method, benefiting from the adaptively supported domain mechanism, maintains a stable convergence behavior with errors remaining at the level of 10^{-5} . This demonstrates that the proposed adaptive mechanism effectively enhances numerical stability and preserves high accuracy even when the standard Local RBF approach begins to lose precision. For both the MQ and IMQ RBFs, the errors remain stable across different N_s , as shown in Table 6.

4.3. 2D Stokes equation

In the steady-state case, the flow field is time-independent. Considering a two-dimensional incompressible flow in a domain Ω , the velocity field is denoted by $\mathbf{u} = (u_1, u_2)^T$ and the pressure field by p . The steady Stokes equations with prescribed Dirichlet boundary conditions are expressed as

$$\begin{cases} -\mu \Delta \mathbf{u} + \nabla p = \mathbf{f}, & \text{in } \Omega, \\ \nabla \cdot \mathbf{u} = 0, & \text{in } \Omega, \\ \mathbf{u} = \mathbf{h}, & \text{on } \partial\Omega, \end{cases} \quad (58)$$

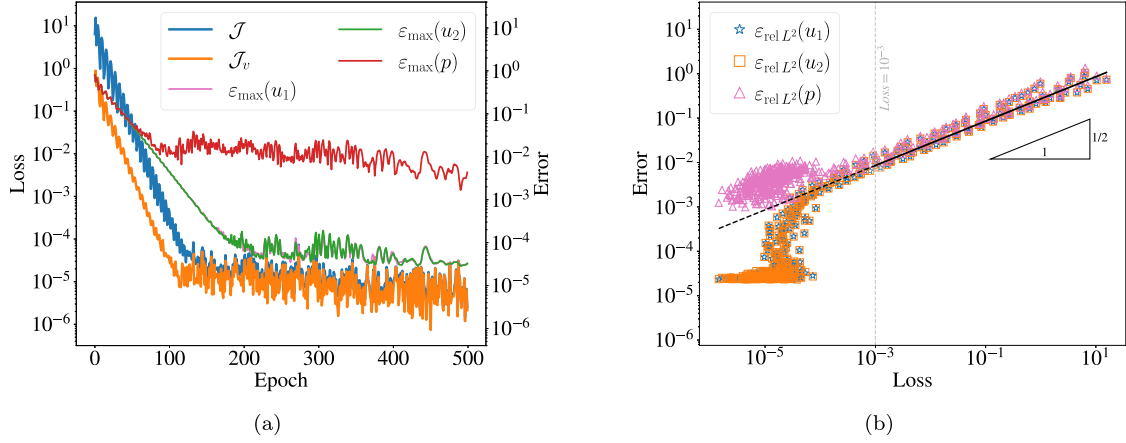


Fig. 12. Training losses and errors for the Local RBF-ASD method in Example 4.3 with $N_s = 30$ and $c_s = 7$: (a) Evolution of $\mathcal{J}$, $\mathcal{J}_v$, and $\epsilon_{\max}$ with epochs; (b) Relation between the relative L^2 error $\epsilon_{\text{rel } L^2}$ and the total loss $\mathcal{J}$.

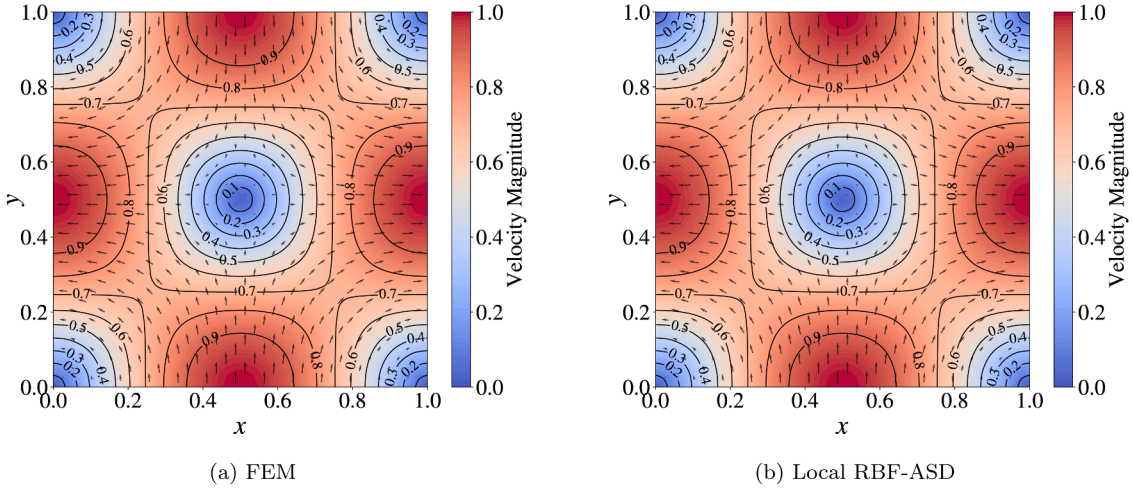


Fig. 13. Velocity field numerical results with $N_s = 30$ and $c_s = 7$ (Example 4.3).

where μ denotes the dynamic viscosity. The computational domain is taken as $\Omega = [0, 1]^2$, and the body force $\mathbf{f}$ is defined as

$$\mathbf{f} = \begin{pmatrix} \cos(\pi x) \sin(\pi y) (\pi - 2\mu\pi^2) \\ \sin(\pi x) \cos(\pi y) (\pi + 2\mu\pi^2) \end{pmatrix}. \quad (59)$$

The corresponding boundary condition is given by

$$\mathbf{h}(x, y) = (-\cos(\pi x) \sin(\pi y), \sin(\pi x) \cos(\pi y)), \quad (60)$$

which ensures that the velocity field satisfies the incompressibility constraint on $\partial\Omega$. Since the pressure p is determined only up to an additive constant, its value is fixed at the lower-left corner of the domain to guarantee uniqueness (Chen et al., 2022). The numerical solution obtained from the ABAQUS finite element solver is adopted as the reference solution for evaluating the accuracy and stability of the proposed Local RBF-ASD method. Collocation points are uniformly distributed over a 40×40 grid, producing $N_{\text{int}} = 1444$ interior and $N_{\text{bd}} = 600$ boundary points.

The variation of the velocity-field errors with respect to the number of local nodes N_s is shown in Fig. 11 for different shape parameters $c_s = \{3, 5, 7\}$. For each chosen c_s , the error curves of u_1 and u_2 obtained by the Local RBF-ASD method consistently remain below those produced by the standard Local RBF scheme. When $c_s = 3$, the errors obtained by the Local RBF-ASD method remain smooth and consistently around the 10^{-4} level, showing only mild variations across different N_s . In contrast, the standard Local RBF method exhibits a wider range of fluctuations, approximately between 1×10^{-2} and 1×10^{-4} , together with

more pronounced oscillations as N_s increases. These results indicate that the adaptively supported domain in the Local RBF-ASD formulation effectively enhances numerical stability. As the shape parameter increases to $c_s = 5$ and $c_s = 7$, the Local RBF-ASD method maintains stable error control, with its maximum error remaining near the order of 10^{-4} . In contrast, the Local RBF method shows distinct instability at several N_s values, where the accuracy deteriorates to approximately 10^{-2} under certain parameter settings, reflecting a loss of numerical robustness.

Based on the error analysis of the pressure solution p for the Stokes equation, as summarized in Table 7, it can be observed that the Local RBF-ASD method consistently outperforms the conventional Local RBF method across the majority of parameter configurations, demonstrating a particularly notable advantage in controlling the maximum error. For instance, at $c_s = 3$, $N_s = 30$, the maximum error of the traditional Local RBF method reaches 2.0338×10^0 , and even at $c_s = 5$, $N_s = 50$, it remains as large as 3.5254×10^{-1} . These results indicate that the traditional approach is highly sensitive to parameter selection. In contrast, the Local RBF-ASD method maintains robust and stable error control throughout the entire parameter space, with its maximum error consistently confined to the order of 10^{-2} , and in most cases reduced further to the 10^{-3} or even 10^{-4} level.

Moreover, among all 24 parameter configurations tested, the Local RBF-ASD method achieved lower relative L^2 errors than the standard Local RBF method in 22 cases. These findings demonstrate that the

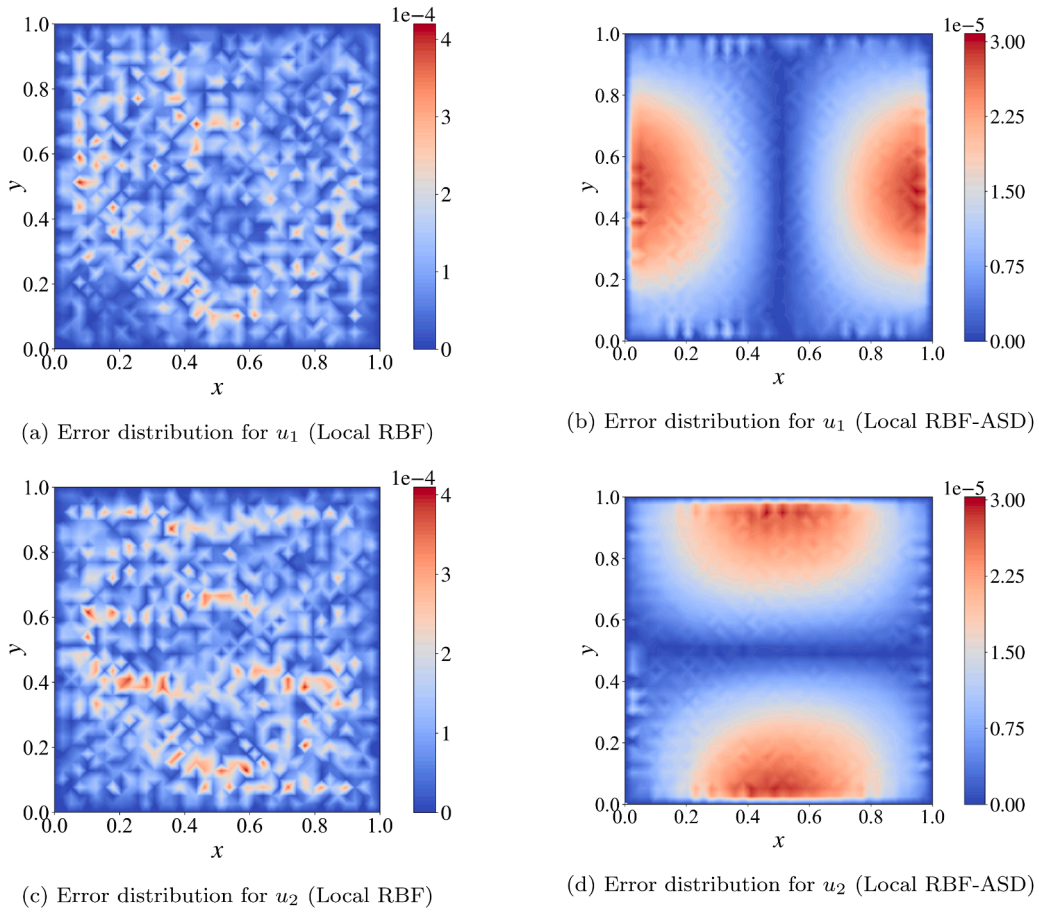


Fig. 14. Velocity field $\mathbf{u}$ error distribution with $N_s = 30$ and $c_s = 7$ (Example 4.3).

adaptive local support domain effectively enhances algorithmic stability and mitigates sensitivity to the selection of local node counts.

As illustrated in Fig. 12a, the training process of the Local RBF-ASD method demonstrates clear and stable convergence under the parameter setting $N_s = 30$, $N = 2044$, and $c_s = 7$. The total loss decreases rapidly from an initial value of 10^1 to below 10^{-3} within the first 100 epochs, and subsequently reaches a steady level around 10^{-5} after approximately 300 epochs. The component losses $\mathcal{J}$ and $\mathcal{J}_v$ follow a similar decay pattern, indicating consistent optimization of the governing and boundary constraints. The pointwise maximum errors of the velocity and pressure fields, $\epsilon_{\max}(\mathbf{u})$ and $\epsilon_{\max}(p)$, exhibit a similar downward trend and eventually stabilize at magnitudes of 10^{-5} for the velocity and 10^{-3} for the pressure, reflecting an accurate and well-conditioned learning process.

It is worth noting that in Fig. 12b, the characteristic trend $\epsilon_{\text{rel } L^2} \propto \mathcal{J}^{1/2}$, as observed in Fig. 5b, does not appear. The reason is that the present problem involves two field variables with distinct physical dimensions. In this case, the total loss function is expressed as Eq. (33). The structural loss term is given by

$$\mathcal{J}_v = \mathcal{J}_{u_1} + \mathcal{J}_{u_2} + \mathcal{J}_p. \quad (61)$$

where $\mathcal{J}_{u_1}$, $\mathcal{J}_{u_2}$, and $\mathcal{J}_p$ denote the loss terms associated with the velocity components u_1 , u_2 , and the pressure p , respectively. When the total loss is relatively large (i.e., $\mathcal{J} \gtrsim 10^{-3}$ in the Fig. 12b), a nearly linear relationship is observed on log-log scales for both the velocity and pressure fields. The corresponding fitted slopes are close to $1/2$, indicating that the relative L^2 errors of both fields are approximately proportional to the square root of the total loss. However, when the loss decreases below 10^{-3} —in particular, when the pressure error falls below 10^{-2} —the structural (physics) loss term $\mathcal{J}_v$ ceases to provide informative gradients due to the limitations of the Local RBF numerical

approximation. In this regime, the condition $\|\nabla \mathcal{J}_p\| \leq \epsilon$ holds, implying that the learning of the pressure field stagnates. Meanwhile, the velocity field continues to improve until its relative error reaches the order of 10^{-5} . In this small-loss regime, the velocity errors exhibit a visible saturation plateau, indicating the onset of a numerically determined error floor.

The computed velocity fields for the two-dimensional Stokes equations are shown in Fig. 13, providing a comparison between the exact and numerical solutions. Fig. 13a displays the reference solution by FEM, while the Fig. 13b shows the numerical solution obtained using the Local RBF-ASD method. Both solutions exhibit agreement in the main flow channel, recirculation regions, and in detailed flow structures, including the position, strength, and rotation direction of vortical patterns. The discrepancies between the two fields are minimal, with no noticeable numerical oscillations or distortions. Furthermore, the vortex cores, influence ranges, and transition zones appear smooth and physically consistent, demonstrating that the Local RBF-ASD method accurately captures the essential flow features.

The error distributions of the velocity components for the two-dimensional Stokes equations, obtained using both the Local RBF-ASD method and the standard Local RBF method, are illustrated in Fig. 14. The comparison reveals that the standard Local RBF method produces errors on the order of 4×10^{-4} for both u_1 and u_2 . In contrast, by incorporating an adaptively supported domain learning strategy, the Local RBF-ASD method reduces the error magnitude to approximately 3×10^{-5} , thereby achieving a substantial improvement in numerical accuracy.

The numerical results for the pressure field of the two-dimensional Stokes equations are compared in Fig. 15. Fig. 15a and c correspond

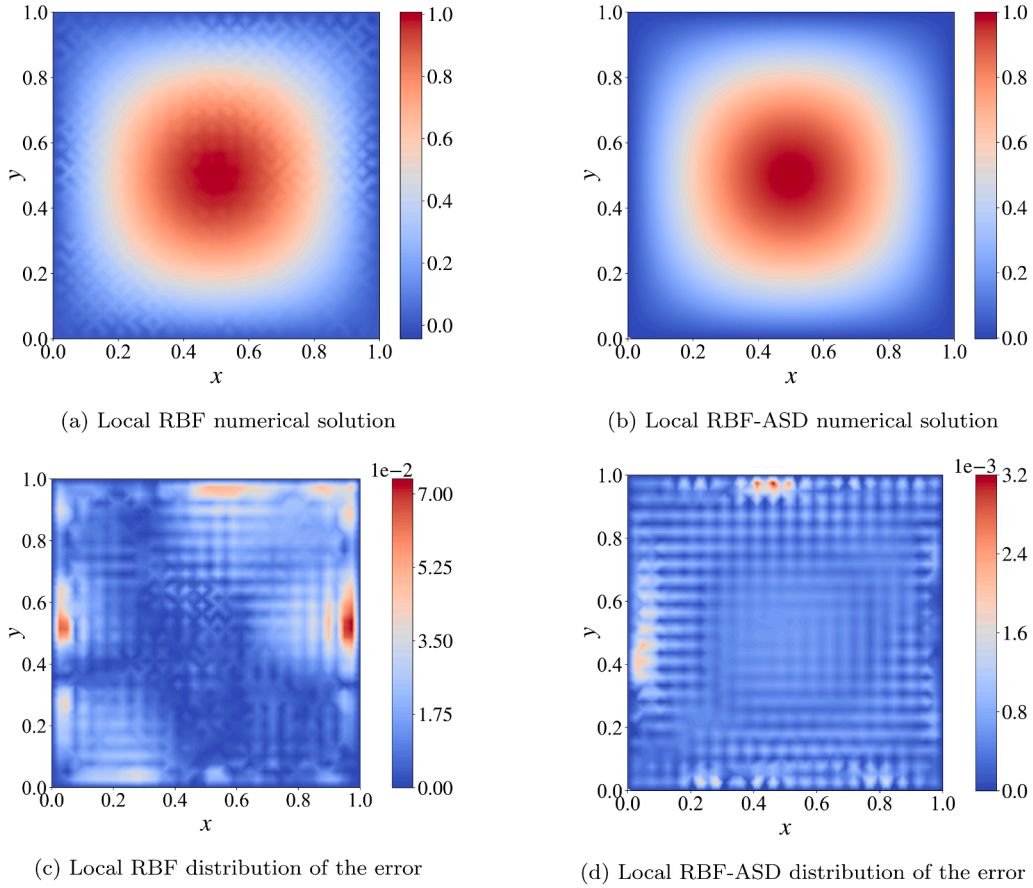


Fig. 15. Pressure field $\mathbf{p}$ numerical results with $N_s = 30$ and $c_s = 7$ (Example 4.3).

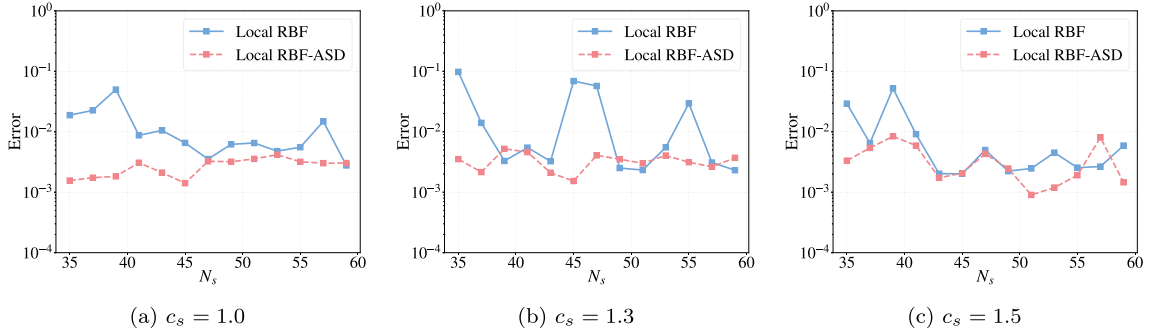


Fig. 16. Variation of the error $\epsilon_{\max}$ with respect to the number of local support nodes N_s under different shape parameters c_s (Example 4.4).

to the Local RBF method, which exhibits pronounced numerical oscillations in the pressure distribution, particularly near geometric boundaries and corner regions. Noticeable deviations from the exact solution are observed, with the maximum error reaching approximately 7.0×10^{-2} . In contrast, the Local RBF-ASD method effectively suppresses these spurious oscillations through its adaptively supported domain mechanism. The resulting pressure field is smooth and well-aligned with the exact solution, with the maximum error reduced to below 3.2×10^{-3} .

4.4. 3D near singular Poisson equation

Consider the following 3D near singular Poisson equation in $\Omega \cup \partial\Omega = [0, 1]^3$:

$$\Delta u(x, y, z) = f(x, y, z), \quad \text{in } \Omega, \quad (62)$$

$$u(x, y, z) = h(x, y, z), \quad \text{on } \partial\Omega, \quad (63)$$

where:

$$f(x, y, z) = \frac{-4a^2 + 3a\sqrt{x^2 + y^2 + z^2} - (x^2 + y^2 + z^2)}{(-a + \sqrt{x^2 + y^2 + z^2})^3}, \quad (64)$$

$$h(x, y, z) = \frac{x^2 + y^2 + z^2}{a - \sqrt{x^2 + y^2 + z^2}}, \quad (65)$$

with the parameter $a = 2.0$. The exact solution to this problem is given by:

$$u^*(x, y, z) = \frac{x^2 + y^2 + z^2}{a - \sqrt{x^2 + y^2 + z^2}}. \quad (66)$$

A uniform $20 \times 20 \times 20$ grid is employed to generate collocation points, yielding $N_{\text{int}} = 5832$ interior points and $N_{\text{bd}} = 4500$ boundary

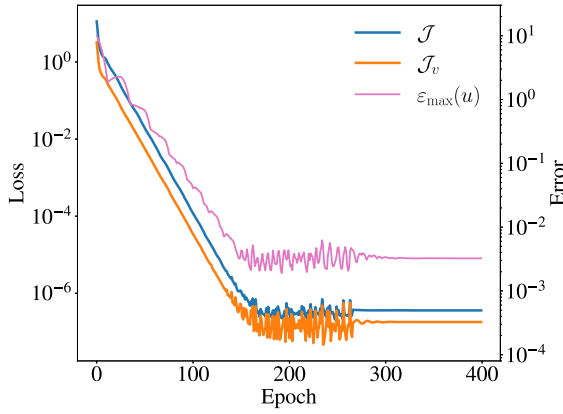


Fig. 17. Loss curve evolution during training with $N_s = 43$ and $c_s = 1$ (Example 4.4).

points. The comparison of the numerical stability with respect to the local support size N_s between the Local RBF and Local RBF-ASD methods is presented in Fig. 16 for shape parameters $c_s \in \{1.0, 1.3, 1.5\}$. When $c_s = 1.0$, the Local RBF-ASD method reduces the error magnitude from the 10^{-2} level of the standard Local RBF method to the 10^{-3} level. Under parameter settings $c_s = 1.3$ and 1.5 , the final error of the Local RBF-ASD method consistently remains below 10^{-2} , whereas the Local RBF method exhibits errors on the order of 10^{-1} at certain collocation points, accompanied by larger error fluctuations. These results demonstrate that, across all parameter configurations, the Local RBF-ASD method achieves higher numerical accuracy and superior stability compared to the conventional Local RBF approach.

The evolution of the loss functions and the velocity error during the iterative training process is illustrated in Fig. 17. The total loss $\mathcal{J}$ and the velocity loss component $\mathcal{J}_v$ both exhibit a rapid decay from an initial magnitude of 10^0 to below 10^{-6} within approximately 200–300 epochs, after which they stabilize, indicating a well-converged training process. Meanwhile, the maximum velocity error $\epsilon_{\max}(u)$ decreases steadily by nearly four orders of magnitude and eventually stabilizes around 10^{-3} . Overall, the loss curves confirm that the Local RBF-ASD method achieves stable and efficient convergence under the configuration $N_s = 43$ and $c_s = 1$.

Numerical results for the near singular Poisson problem described in Example 4.4 are presented in Fig. 18. Fig. 18a and b present the scalar-field solutions computed by the Local RBF and Local RBF-ASD methods. To reveal the interior structure of the three-dimensional solution, a cut-away surface is introduced. The slicing plane is chosen to pass through the three points $(1, 1, 1)$, $(1, 0, 0)$, and $(0, 1, 0)$, yielding the planar equation $x + y - z = 1$. This cut-plane removes a portion of the exterior domain and exposes the internal variation of the solution, thereby making the sharp gradients near the nearly singular region visible and enabling direct visual comparison of the two methods. Both approaches successfully capture the pronounced gradients induced near the nearly singular region, with the solution range varying from approximately 0 to 11.2. Fig. 18c and d present the pointwise error distributions evaluated on several horizontal slices $z \in \{0.05, 0.275, 0.5, 0.725, 0.95\}$. For the standard Local RBF approach, the maximum numerical error is approximately 1.25×10^{-2} , and several localized error peaks emerge in the solution. By contrast, the Local RBF-ASD formulation reduces the magnitude of the error substantially, with the maximum around 2.5×10^{-3} , and yields a more uniform and less oscillatory error pattern across all slices. These results collectively demonstrate the efficacy of the

adaptively supported domain enhancement approach in providing improved accuracy and enhanced numerical stability for this challenging three-dimensional near-singular problem.

4.5. 2D convection–diffusion equation

We consider the two-dimensional convection–diffusion equation on the domain $\Omega = [0, 1] \times [0, 1]$:

$$\begin{cases} -\epsilon \nabla^2 u(x, y) + \beta_x \frac{\partial u(x, y)}{\partial x} + \beta_y \frac{\partial u(x, y)}{\partial y} = f(x, y), & (x, y) \in \Omega, \\ u(x, y) = h(x, y), & (x, y) \in \partial\Omega. \end{cases} \quad (67)$$

Here, $\epsilon > 0$ is the diffusion coefficient, and β_x and β_y are the convection coefficients in the x - and y -directions, respectively. Dirichlet boundary conditions are imposed.

The exact solution is prescribed as

$$u(x, y) = \frac{1}{2} \left[1 + \tanh \left(\frac{x - x_0 - \alpha \sin(2\pi y)}{\delta} \right) \right], \quad (68)$$

where $x_0 = 0.5$, $\alpha = 0.2$, and $\delta = 3\epsilon$. The source term f and boundary data h are obtained by substituting (68) into (67). In this example, we set $\epsilon = 0.01$ and $(\beta_x, \beta_y) = (1.0, 0.5)$, giving $\delta = 3 * \epsilon$. The small diffusion coefficient produces a thin curved internal transition layer and makes the problem strongly convection-dominated, thereby providing a challenging test for the stability of the local support construction.

To assess the performance of Local RBF and Local RBF-ASD for this sharp-layer problem, a 40×40 uniform tensor grid is generated, yielding $N_{\text{int}} = 1444$ interior collocation points after excluding the boundary nodes, together with $N_{\text{bd}} = 200$ boundary points. IMQ radial basis functions are employed in both methods.

To investigate the sensitivity to the local support size, the maximum pointwise error $\epsilon_{\max}$ is evaluated for N_s ranging from 20 to 60, with the shape parameter chosen as $c_s \in \{4, 5, 6\}$. The corresponding results are presented in Fig. 19. A representative numerical solution and its error distribution for $N_s = 45$ and $c_s = 4$ are further shown in Fig. 20.

As shown in Fig. 19, the error of Local RBF varies considerably with N_s and may exhibit severe amplification for certain combinations of N_s and c_s . In contrast, Local RBF-ASD maintains the maximum error within a substantially narrower range across the tested support sizes and shape parameters. This demonstrates that the adaptive support-domain mechanism significantly reduces the sensitivity of the numerical solution to the prescribed value of N_s .

Fig. 20 further shows that both methods capture the thin curved transition layer, while the error produced by Local RBF-ASD remains more localized and has a smaller magnitude. These results confirm the improved stability of Local RBF-ASD for convection-dominated problems containing sharp internal layers.

Table 8 shows that the conventional Local RBF method requires between 0.23 and 1.308 s, whereas Local RBF-ASD requires between 168.72 and 560.43 s on the same CPU platform. Therefore, the current Local RBF-ASD implementation is approximately 353–1086 times more expensive than a single conventional Local RBF solve for the tested values of N_s .

This computational overhead is expected because Local RBF constructs and solves the discrete system once, whereas Local RBF-ASD repeatedly evaluates the gated local systems and updates the solution and gating parameters over multiple optimization epochs. Therefore, the proposed method does not provide a runtime advantage. Instead, the additional computational cost is exchanged for reduced sensitivity to N_s and improved numerical robustness.

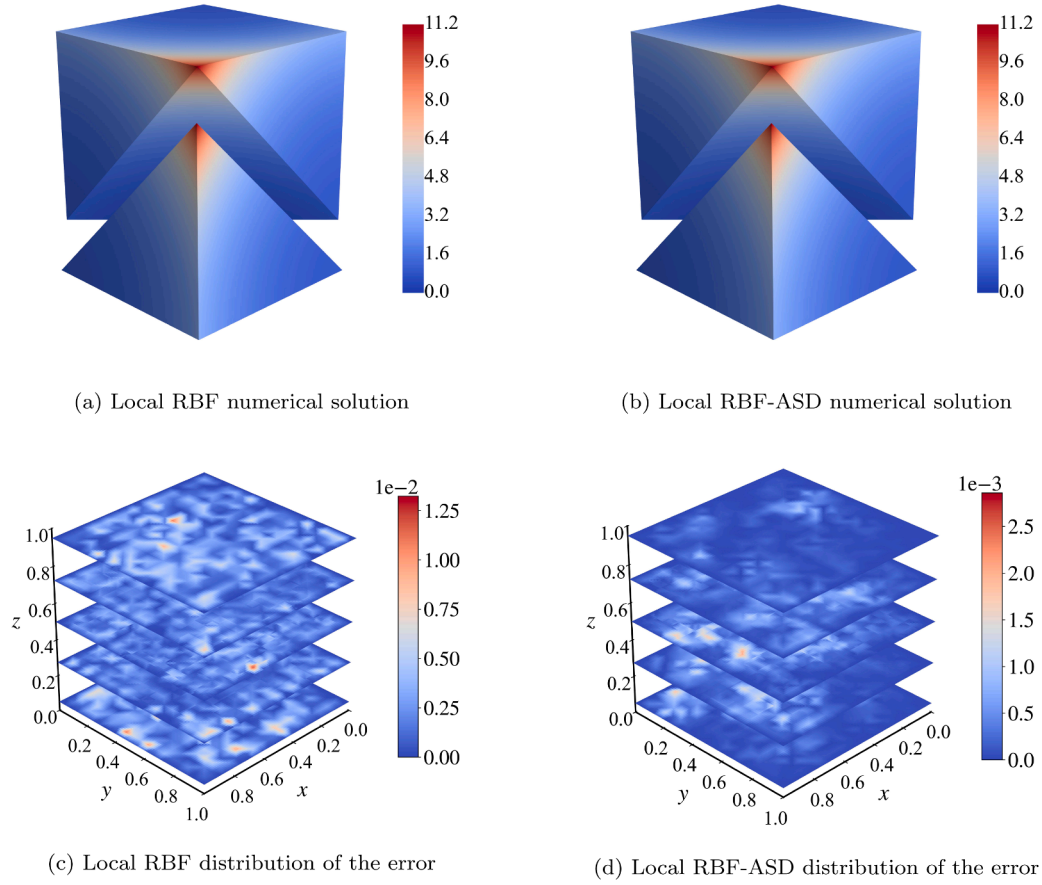


Fig. 18. Numerical results for the 3D near singular poisson equation with $N_s = 43$ and $c_s = 1$ (Example 4.4).

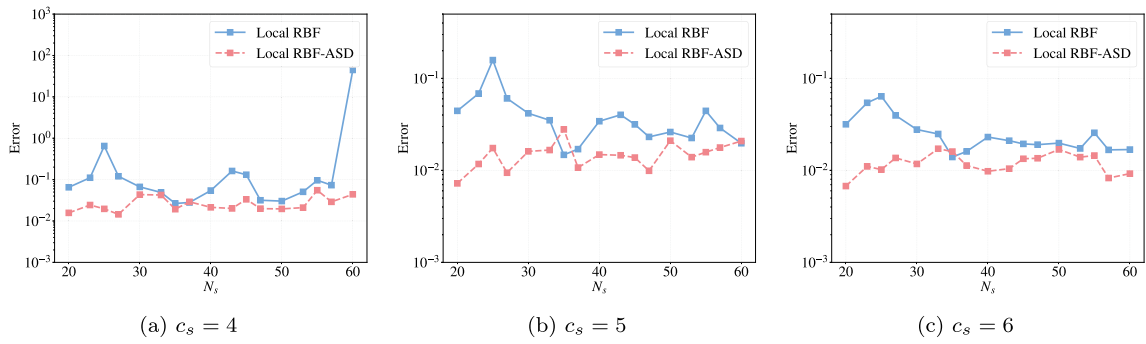


Fig. 19. Variation of the error $\epsilon_{\max}$ with respect to the number of local support nodes N_s under different shape parameters c_s (Example 4.5).

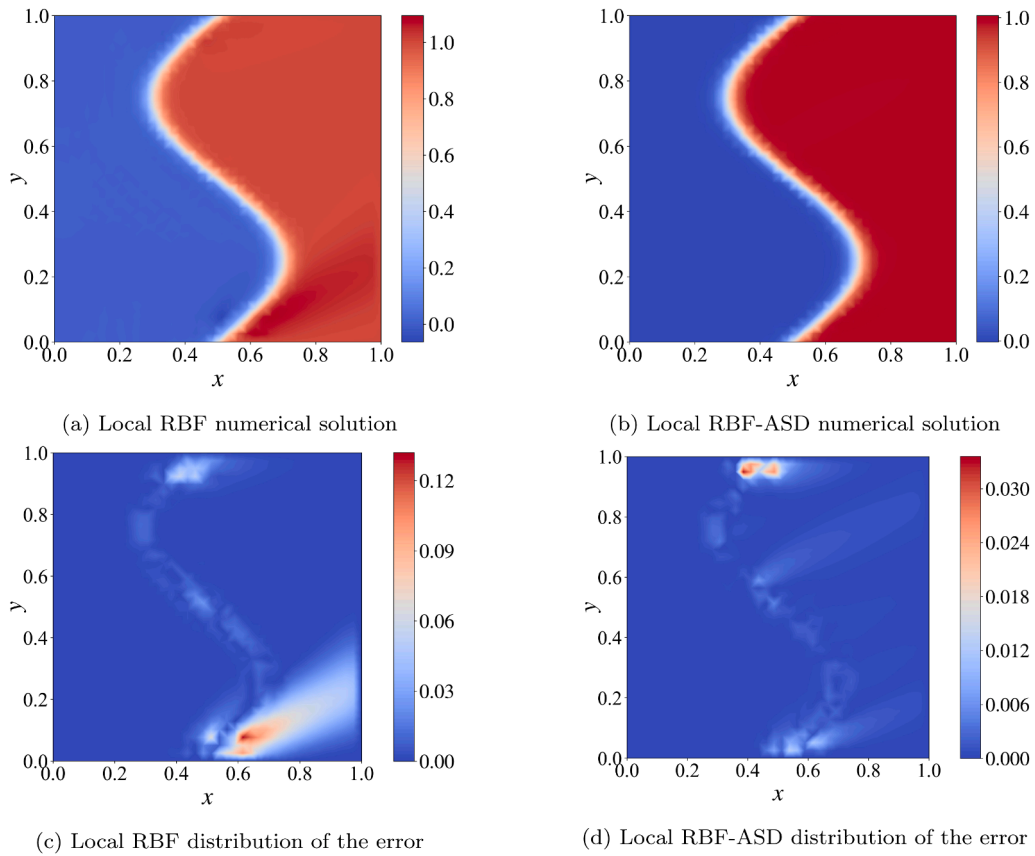


Fig. 20. Numerical results with $N_s = 45$ and $c_s = 4$ (Example 4.5).

5. Conclusion

In this work, we have developed the Local RBF-ASD method that addresses a fundamental challenge in local radial basis function methods: the sensitivity of numerical accuracy to the choice of local support domain size. By introducing learnable gating parameters $\{b_R^{(s)}\}$ within a differentiable training neural network framework, the proposed method adaptively determines effective support domains for each collocation point, thereby eliminating the need for heuristic or trial-and-error parameter selection that has long plagued conventional Local RBF approaches. The composite loss function, combining PDEs residuals with boundary constraints, enables joint optimization of learnable parameters and unknown field quantities while preserving the sparse structure essential for computational efficiency.

Extensive numerical experiments on benchmark problems demonstrate the effectiveness and robustness of the proposed approach. In particular, the results indicate that the method can reduce sensitivity to the choice of support domain sizes. Overall, these findings suggest that Local RBF-ASD provides a reliable and accurate neural network framework that alleviates the parameter sensitivity inherent in traditional Local RBF methods. Beyond its specific formulation, the proposed Local RBF-ASD method provides a general paradigm. The core idea of embedding differentiable, parameterized gating mechanisms into local approximation schemes can be readily extended to other localized methods and discretization schemes, offering a unified strategy for mitigating parameter sensitivity and improving solution quality. Future research directions include extending the adaptive framework to time-dependent problems, investigating its application to nonlinear PDEs, and exploring more sophisticated gating mechanisms to further enhance the method's approximation capabilities.

CRedit authorship contribution statement

Zhenzhou Fan: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization; **Min Lei:** Writing – review & editing, Supervision; **Ruiping Niu:** Writing – review & editing, Supervision, Funding acquisition; **Letian Wang:** Methodology, Investigation; **Hongen Jia:** Supervision, Funding acquisition, Formal analysis; **Peiyao Liu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the Regional Innovation and Development Joint Fund of the National Natural Science Foundation of China (Grant No. U25A20812).

References

- Bai, J., Liu, G.-R., Gupta, A., Alzubaidi, L., Feng, X.-Q., & Gu, Y. (2023). Physics-informed radial basis network (PIRBN): A local approximating neural network for solving nonlinear partial differential equations. *Computer Methods in Applied Mechanics and Engineering*, 415, 116290.
- Belytschko, T., Krongauz, Y., Organ, D., Fleming, M., & Krysl, P. (1996). Meshless methods: An overview and recent developments. *Computer Methods in Applied Mechanics and Engineering*, 139(1–4), 3–47.
- Belytschko, T., Lu, Y. Y., & Gu, L. (1994). Element-free Galerkin methods. *International Journal for Numerical Methods in Engineering*, 37(2), 229–256.

- Chandhini, G., & Sanyasiraju, Y. (2010). Local RBF-FD solutions for steady convection–diffusion problems. *International Journal for Numerical Methods in Engineering*, 72(3), 352–378.
- Chen, J., Chi, X., Yang, Z. et al. (2022). Bridging traditional and machine learning-based algorithms for solving PDEs: The random feature method. *Journal of Machine Learning*, 1(3), 268–298.
- Chen, W., & Tanaka, M. (2002). A meshless, integration-free, and boundary-only RBF technique. *Computers & Mathematics with Applications*, 43(3–5), 379–391.
- Chen, Y., Cheng, J., Li, T., & Miao, Y. (2025). A learning based numerical method for Helmholtz equations with high frequency. *Journal of Computational Physics*, 520, 113478.
- Gingold, R. A., & Monaghan, J. J. (1977). Smoothed particle hydrodynamics: Theory and application to non-spherical stars. *Monthly Notices of the Royal Astronomical Society*, 181(3), 375–389.
- Han, L., Lei, M., Wu, C. T., Niu, R. P., Jia, H. E., & Wang, H. B. (2025). Neural network-driven adaptive parameter selection for the local method of fundamental solutions. *Engineering Analysis with Boundary Elements*, 178, 106247.
- Kansa, E. J. (1990a). Multiquadrics—a scattered data approximation scheme with applications to computational fluid-dynamics—I: Surface approximations and partial derivative estimates. *Computers & Mathematics with Applications*, 19(8-9), 127–145.
- Kansa, E. J. (1990b). Multiquadrics—a scattered data approximation scheme with applications to computational fluid-dynamics—II: Solutions to parabolic, hyperbolic and elliptic partial differential equations. *Computers & Mathematics with Applications*, 19(8-9), 147–161.
- Kundu, P. K., Cohen, I. M., Dowling, D. R., & Capecelatro, J. (2024). *Fluid mechanics*. Elsevier.
- Kupradze, V. D., & Aleksidze, M. A. (1964). The method of functional equations for the approximate solution of certain boundary value problems. *USSR Computational Mathematics and Mathematical Physics*, 4(4), 82–126.
- LeVeque, R. J. (2007). Finite difference methods for ordinary and partial differential equations: Steady-state and time-dependent problems. SIAM.
- Li, Y., Liu, D., Yin, Z., Chen, Y., & Meng, J. (2023). Adaptive selection strategy of shape parameters for LRBF for solving partial differential equations. *Applied Mathematics and Computation*, 440, 127645.
- Lucy, L. B. (1977). A numerical approach to the testing of the fission hypothesis. *Astronomical Journal*, 82, 1013–1024.
- Marin, L., & Lesnic, D. (2007). The method of fundamental solutions for nonlinear functionally graded materials. *International Journal of Solids and Structures*, 44(21), 6878–6890.
- Mathon, R., & Johnston, R. L. (1977). The approximate solution of elliptic boundary-value problems by fundamental solutions. *SIAM Journal on Numerical Analysis*, 14(4), 638–650.
- Nayroles, B., Touzot, G., & Villon, P. (1992). Generalizing the finite element method: Diffuse approximation and diffuse elements. *Computational Mechanics*, 10(5), 307–318.
- Qian, Y., Zhang, Y., Huang, Y., & Dong, S. (2023). Physics-informed neural networks for approximating dynamic (Hyperbolic) PDEs of second order in time: Error analysis and algorithms. *Journal of Computational Physics*, 495, 112527.
- Qiao, Y., Zhai, S., & Feng, X. (2017). RBF-FD method for the high dimensional time fractional convection–diffusion equation. *International Communications in Heat and Mass Transfer*, 89, 230–240.
- Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378, 686–707.
- Rippa, S. (1999). An algorithm for selecting a good value for the parameter C in radial basis function interpolation. *Advances in Computational Mathematics*, 11(2-3), 193–210.
- Sarra, S. A. (2023). Local radial basis function methods: Comparison, improvements, and implementation. *Journal of Applied Mathematics and Physics*, 11(12), 3867–3886.
- Sawaf, B., Ozisik, M. N., & Jarny, Y. (1995). An inverse analysis to estimate linearly temperature dependent thermal conductivity components and heat capacity of an orthotropic medium. *International Journal of Heat and Mass Transfer*, 38(16), 3005–3010.
- Shu, C., Ding, H., Chen, H. Q., & Wang, T. G. (2005). An upwind local RBF-DQ method for simulation of inviscid compressible flows. *Computer Methods in Applied Mechanics and Engineering*, 194(18–20), 2001–2017.
- Versteeg, H. K. (2007). An introduction to computational fluid dynamics the finite volume method, 2/E. Pearson Education India.
- Wang, F., Li, X., Liu, H., Qiu, L., & Yue, X. (2025). An adaptive method of fundamental solutions using physics-informed neural networks. *Engineering Analysis with Boundary Elements*, 178, 106295.
- Wendland, H. (2005). Scattered data approximation (vol. 17). Cambridge University Press.
- Zhang, H. H., Genton, M. G., & Liu, P. (2004). Compactly supported radial basis function kernels. Technical Report North Carolina State University. Dept. of Statistics.
- Zhi-Qin, J., Xu, J., Tao, L., Yanyang, X., & Zheng, M. (2020). Frequency principle: Fourier analysis sheds light on deep neural networks. *Communications in Computational Physics*, 28(5), 1746–1767.
- Zienkiewicz, O. C., Taylor, R. L., & Zhu, J. Z. (2005). *The finite element method: Its basis and fundamentals*. (6th ed.). Elsevier.